GOSIM®

GLOBAL OPEN-SOURCE INNOVATION MEETUP

GOSIM Shenzhen 2026

10月16-17日 October 16-17　　深圳·南山伊敦酒店 Aden Hotel, Shenzhen

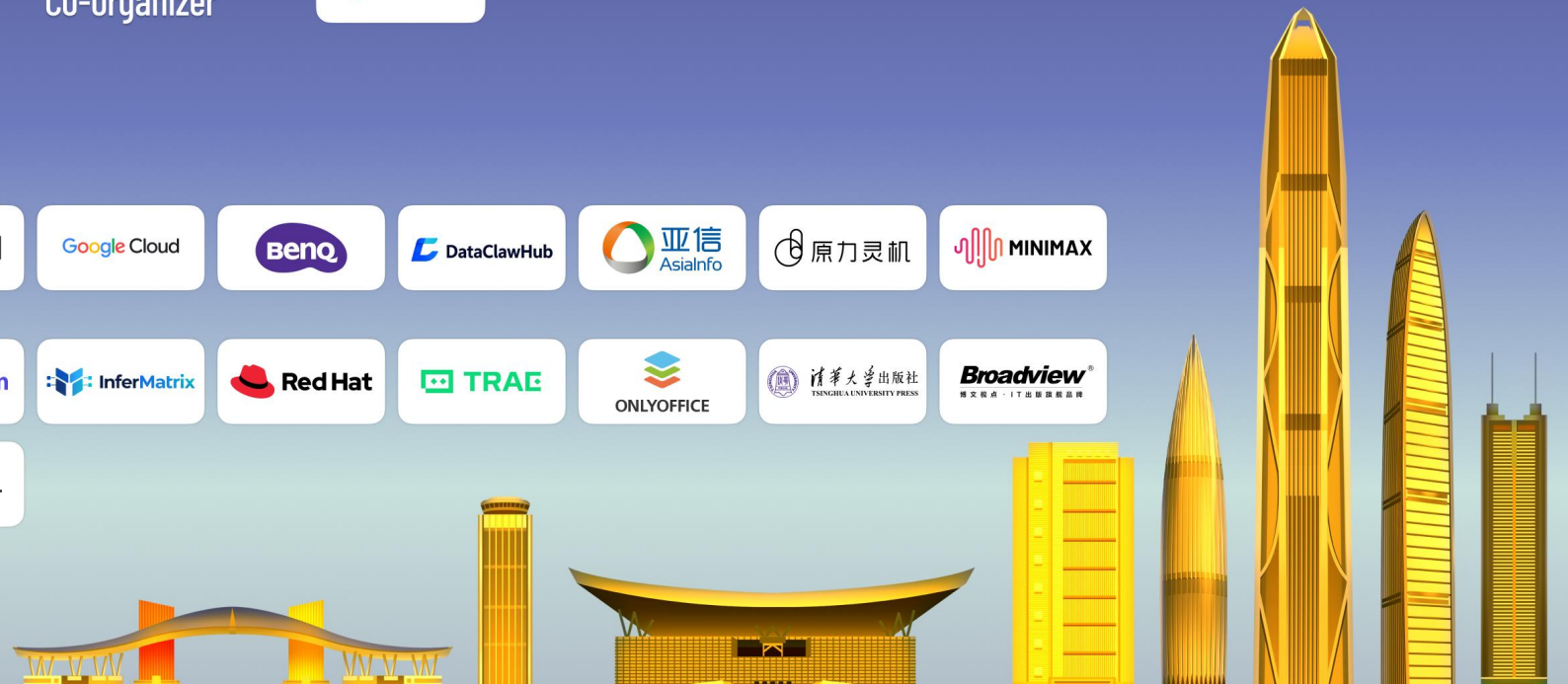
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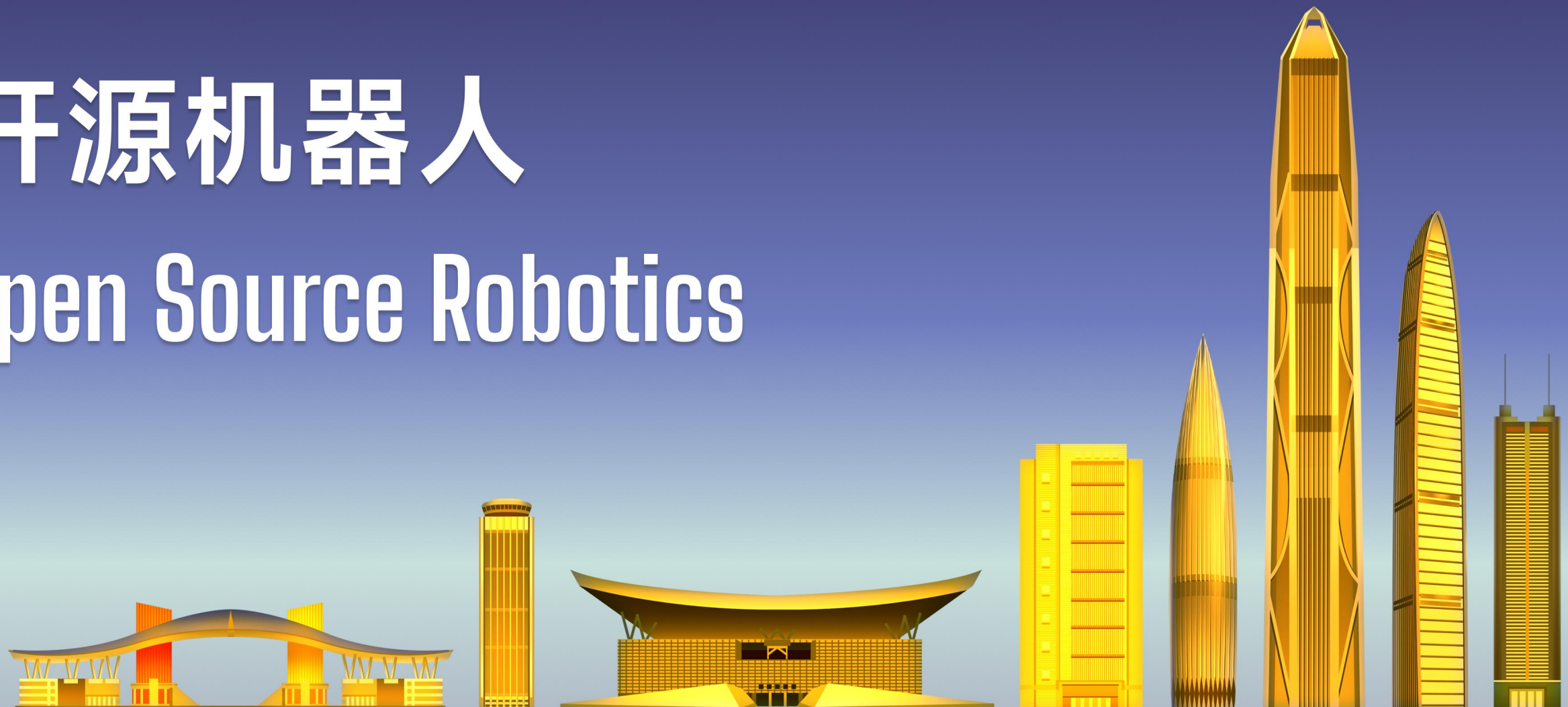


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开源机器人

Open Source Robotics



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Topic

“World Model for All”：迈向生成、导航与控制的统一

"World Model for All": Towards Unified Generation, Navigation and Control



王言治 Yanzhi Wang

美国东北大学教授

Professor, Northeastern University



Physical AI Research (PAIR) Center

World Model for Representation, Generation and Embodied AI

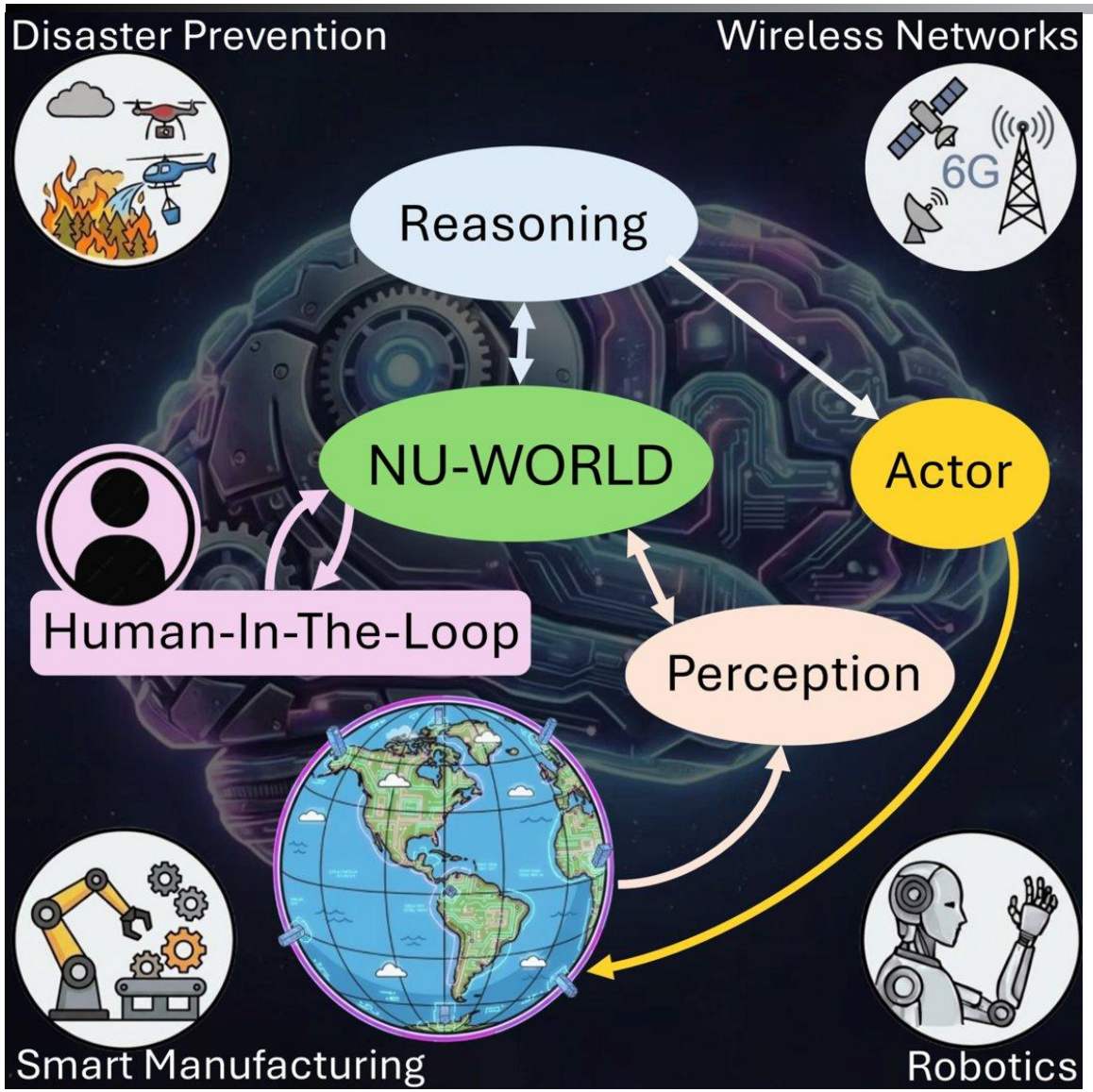
Yanzhi Wang

Professor and Faculty Fellow
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October 12, 2026

Northeastern University
College of Engineering

Towards Open-Source World Model in Academia



- Senses and learns real-world knowledge.
- Predicts and generates real-world scenes.
- Reasons and controls according to physical laws.
- Acts robustly with human-in-the-loop.

Currently there is no open-source, physically grounded world model from academia

Categories of World Model

#1. 3D scene generation (World Labs)

#2. JEPA-like representation learning and prediction

#3. Controlled real-world scene generation (Nvidia Cosmos, Wan 2.2, etc.)

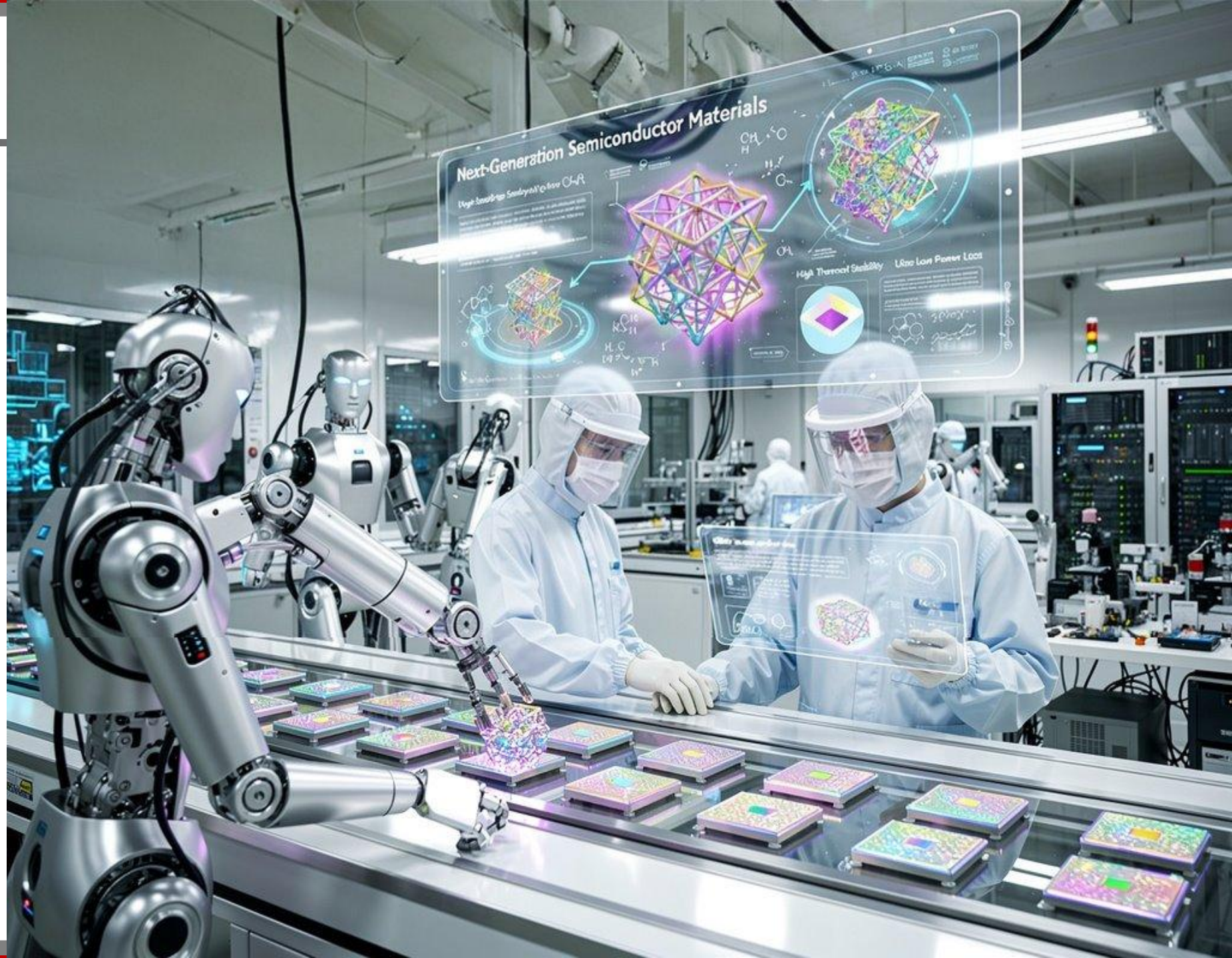
#4. Human interactive scene generation (Google Genie 3, Lingbot-World, etc.)



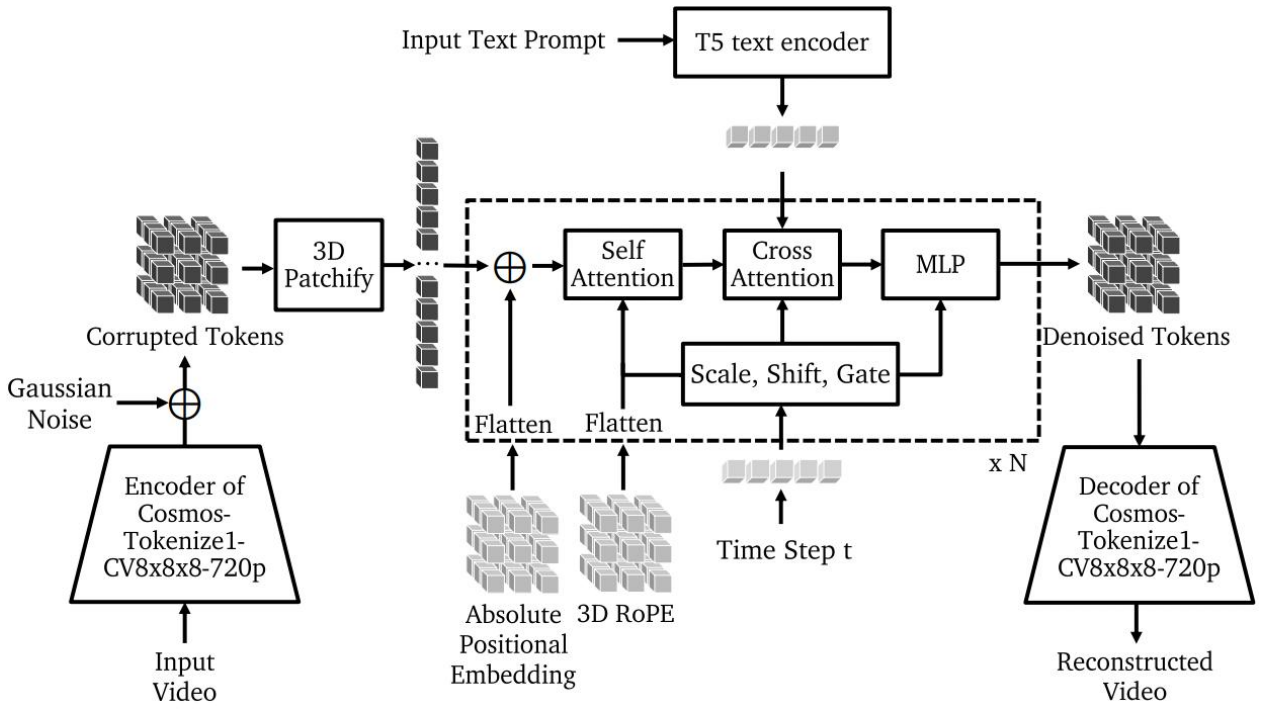
Application in Embodied AI & Manufacturing

As a key application of world model

World model can help with control, human interaction, planning, data generation, 3D reconstruction, etc.



Nvidia Cosmos: Structure & Limitation

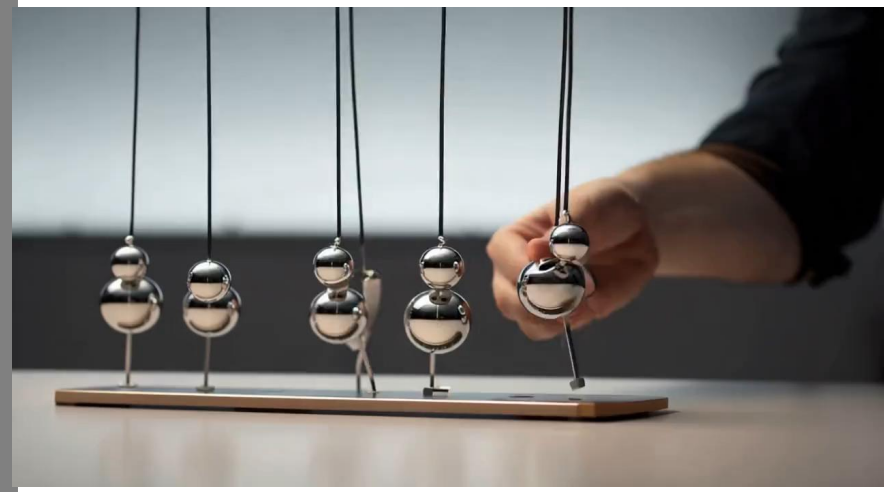




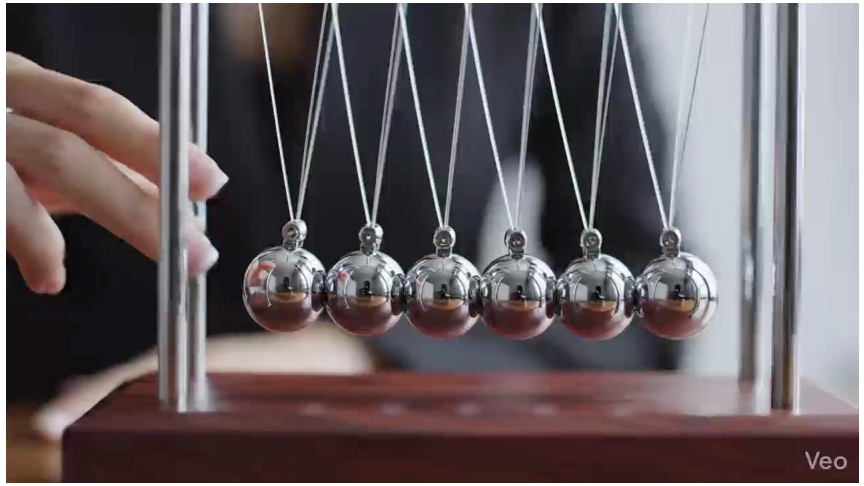
Major Limitations:

- The video encoder is quite weak (cannot be used for reconstruction)
- Short-term generation (5-10s)
- The physics awareness is limited – inherent through physical scene training

Difficulty in Physical Awareness



NVIDIA Cosmos

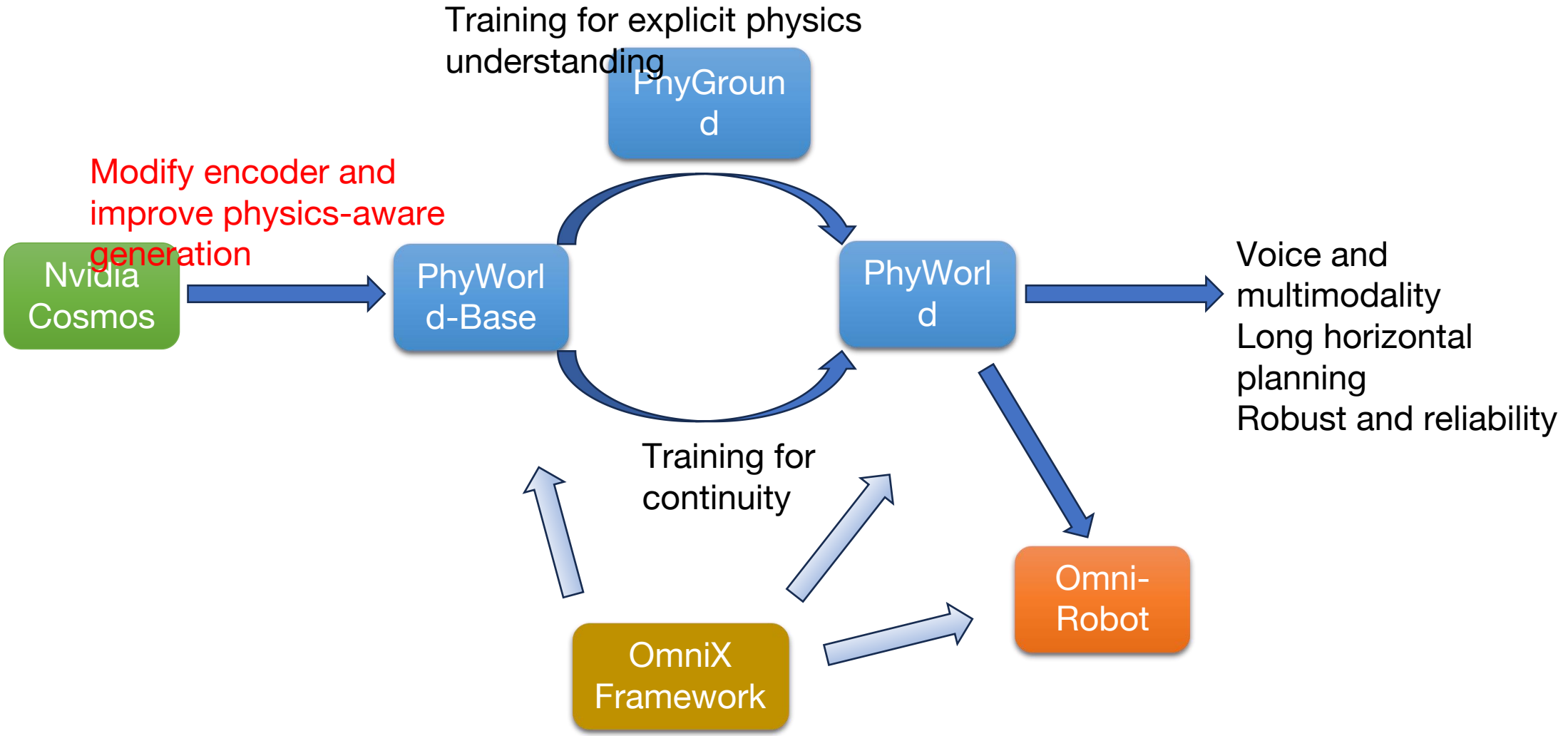


Google Veo

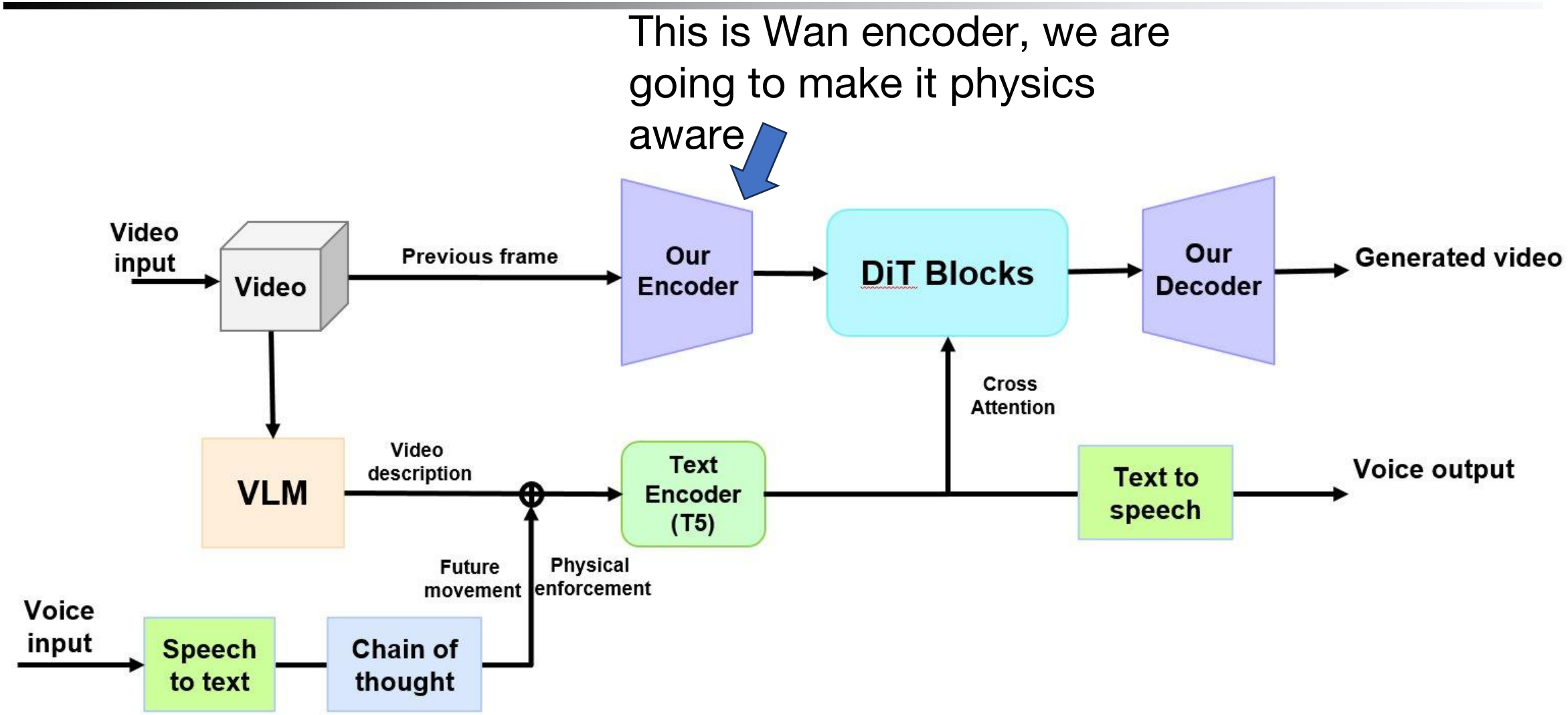


OpenAI Sora

Our Innovations and Framework



World Model Base (PhyWorld-Base): Structure



Making Wan2.2 Encoder Physics-Aware

- Extend from image2video to video2video
- Extend the mask from first frame to first a few frames
- Train with video continuation datasets

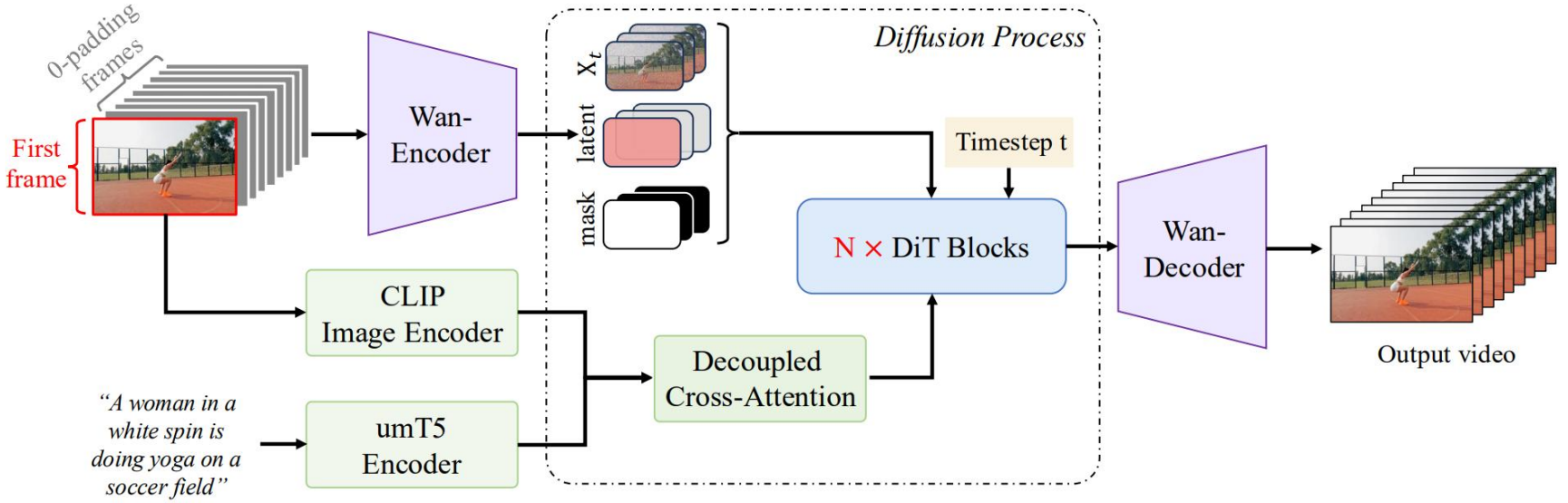
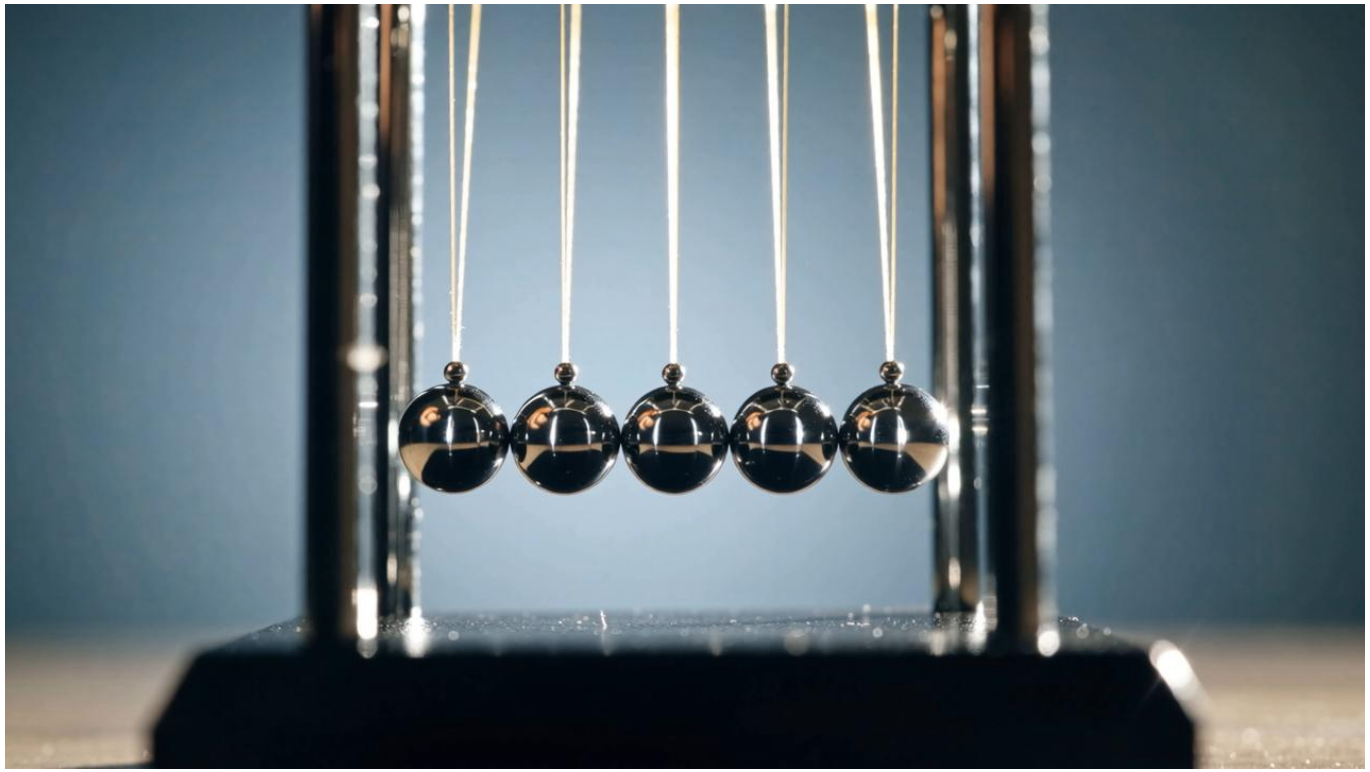


Figure 18: Wan-I2V model framework. The mask mechanism enables the model to selectively focus on input frames such that the proposed framework demonstrates compatibility with video continuation and first-last frame transformation tasks.

Quality of PhyWorld-Base



Quality of PhyWorld-Base



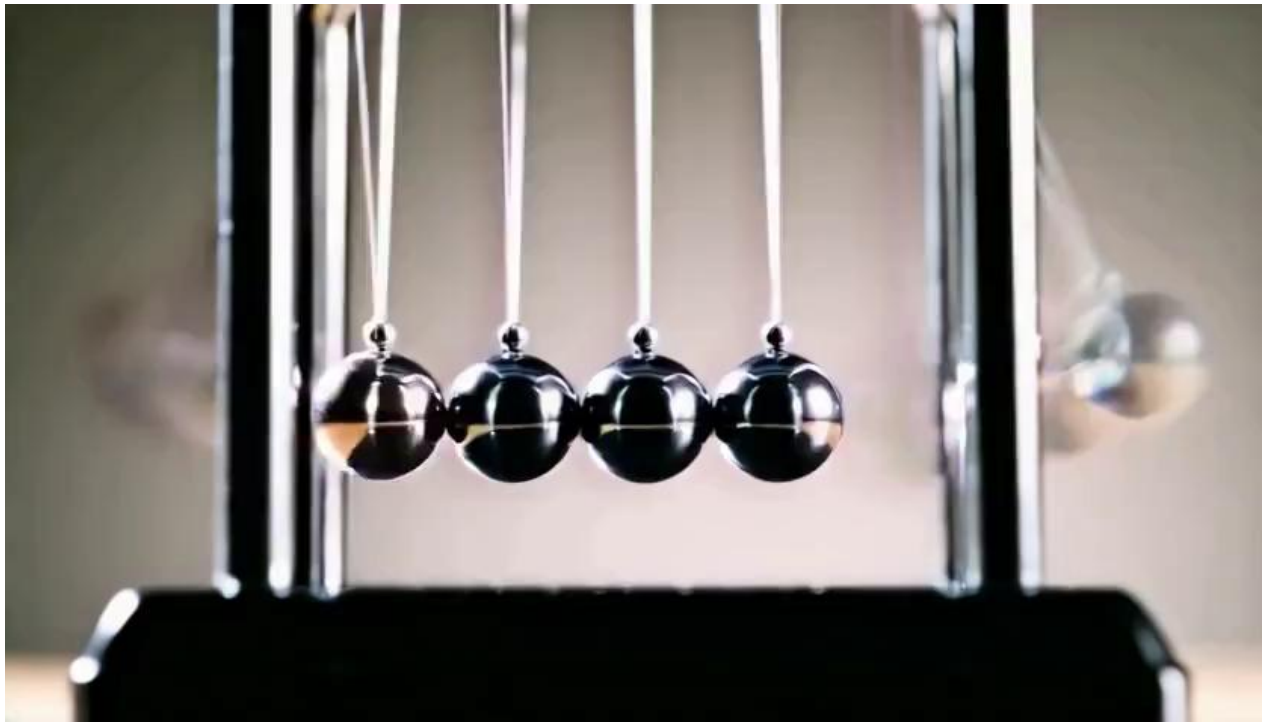
NVIDIA Cosmos: not realistic



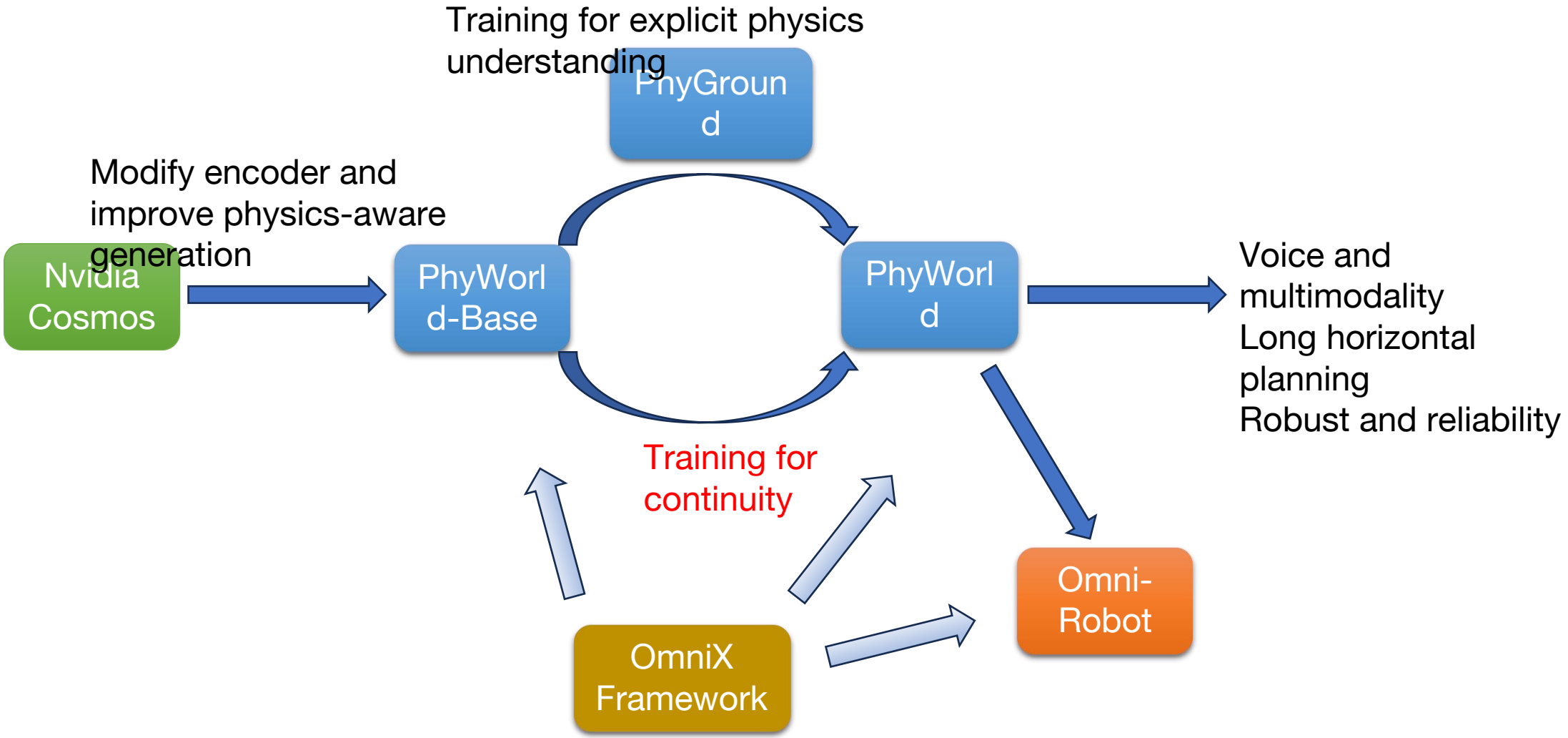
PhyWorld-Base: more realistic

Limitation of PhyWorld-Base

- Limited capability in long horizontal generation (e.g., color shift, speed change)
- Still limitation in physics-awareness, especially in physics understanding



Our Innovations and Framework



World Model Step I (Improve Continuity): Prior Work & Limitations

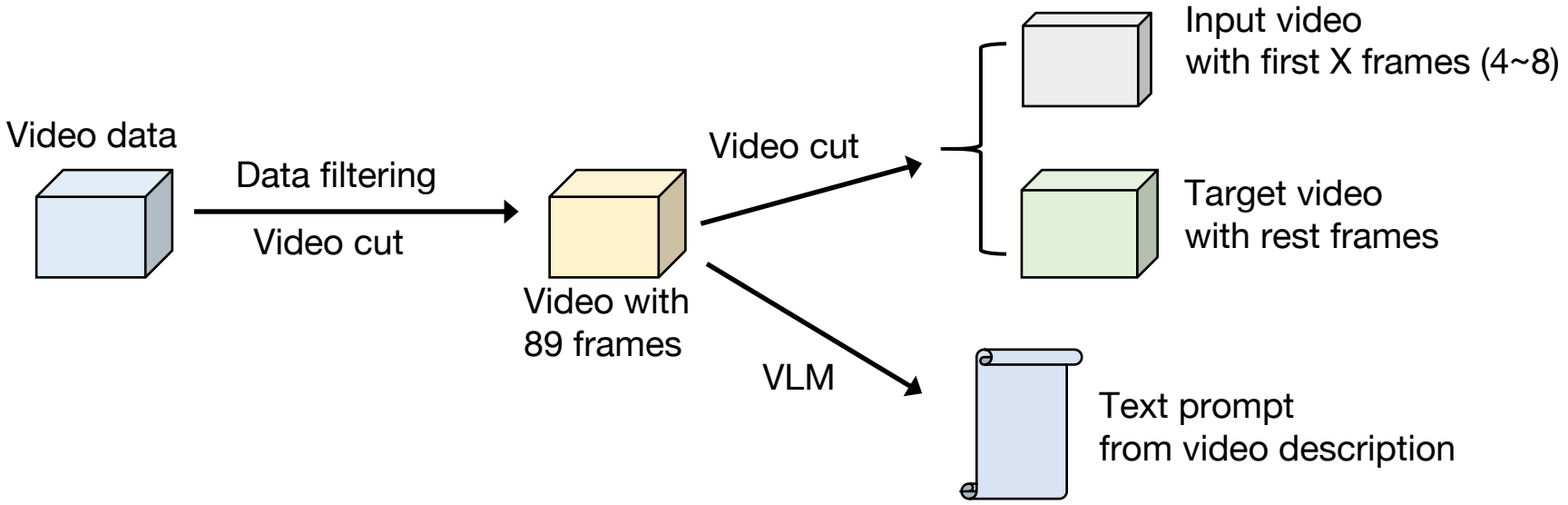
- Prior work are mostly Self Forcing or LongCat based – Focusing on ultra-long video generation
  - Including our series of work such as OmniMem
- Limitations
  - Small video generation models (large memory usage during training)
  - Consistency problem and not physics aware





PhyWorld Step I (Improve Continuity): Data Pipeline

- Data Pipeline Output: input video, target video, text prompt
- Objective: target video continues the input video following text prompt



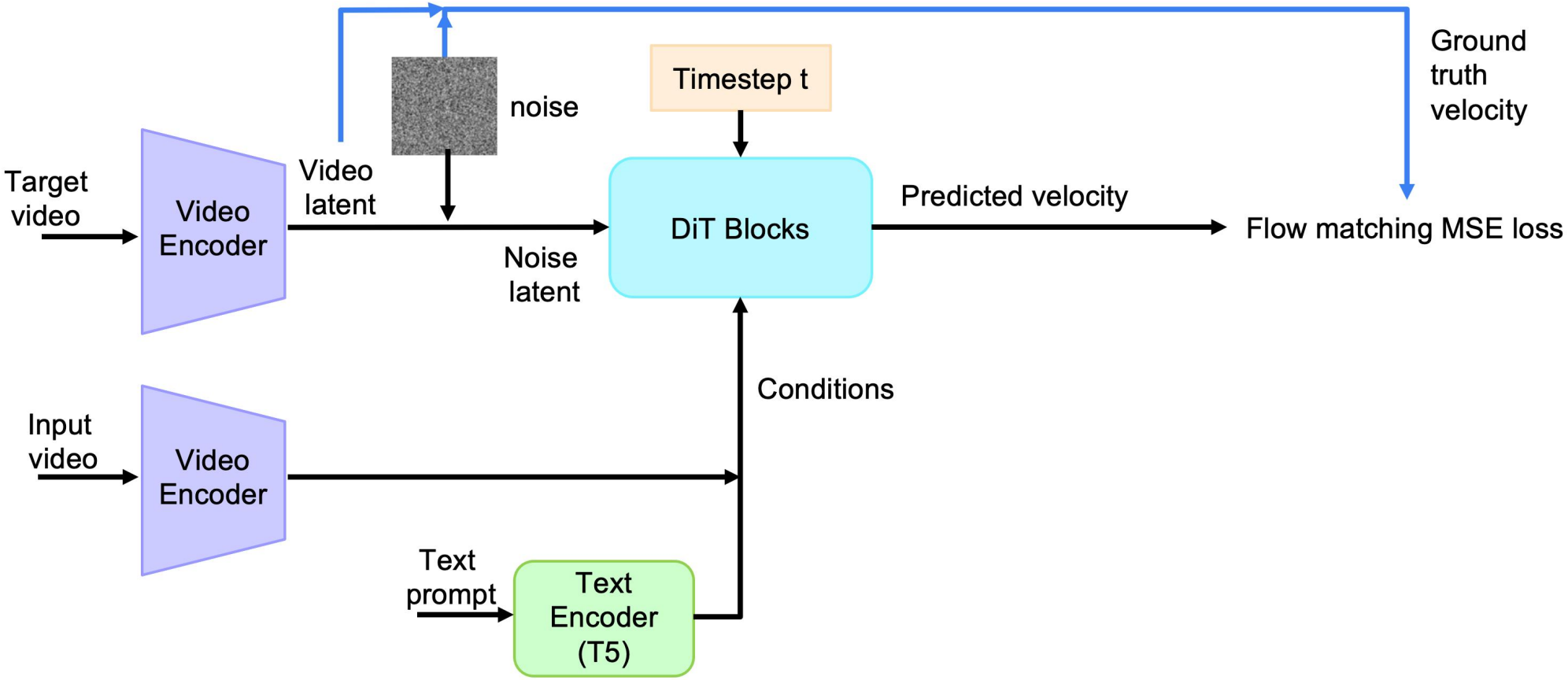
PhyWorld Step I: Datasets and Data Filtering

Category	Dataset Name	1. Frame Spec (Res / FPS)	2. Duration / Count	3. Size (Storage)	4. Text Label?	link
Auto Driving	NVIDIA PhysicalAI	1920x1080 @ 30fps	Clip: 20s Total: 1727 hrs	Total: ~1000TB, Open: 70TB	No(LiDAR, Radar, Parquet Meta)	
	BDD100K	1280x720 @ 30fps	Clip: 40s Total: 100K videos	1.8 TB	No(GPS/IMU, Box, Seg @ keyframe)	
	NVIDIA Cosmos-Drive	Synthetic: 121 frames ~12 fps	Clip: 10s Total: ~87K clips	~3 TB	No(Synthetic Labels, Weather)	
	Waymo Open	1920x1280 .10–20 Hz,	Clip: 20s Total: 570+ hrs (Motion)	Compressed (TFRecord)	No(3D Points, Tracks, Maps)	
Robotics	AgiBotWorld-Beta	N/A (MP4 Avail)	Total: 2976 hrs1M videos	48 TB	Yes(MP4, Actions) "action_text": "Place onion into the	
	DROID	720p@60 / 1080p@30 stereo	Total: 350 hrs76K videos	Raw: 8.7 TB, RLDS: 1.7 TB	No(MP4, Actions)	
	BridgeData V2	640x480 (VGA)5 Hz,	Clip: ~7-8s Total: 60K videos	Small (TFDS format)	No(Actions/State)	
General	Panda-70M	1280x720	Clip: ~8.5s Total: 167K hrs	~36 TB	Yes(Video Captions/Descriptions)	
	DL3DV-10K	3840x2160 (4K) @ 60fps	Clip: ~69.5s Total: 10K videos	Large (Raw frames)	No(Camera Pose/COLMAP)	
	RealEstate-10K	Variable (YouTube)	Total: 10K videos	Text only (URLs)	No(Pose/Timestamp only)	
	action100m-preview	Variable (YouTube)	1.2M clips （14.6 years）	40GB in hf, all 400GB in github	yes	https://huggingface.c
	OpenVid-1M	Variable (YouTube) fps 25-30	1M clips, inlcuding 0.4M clips 1080	4.5TB	yes	https://huggingfa http

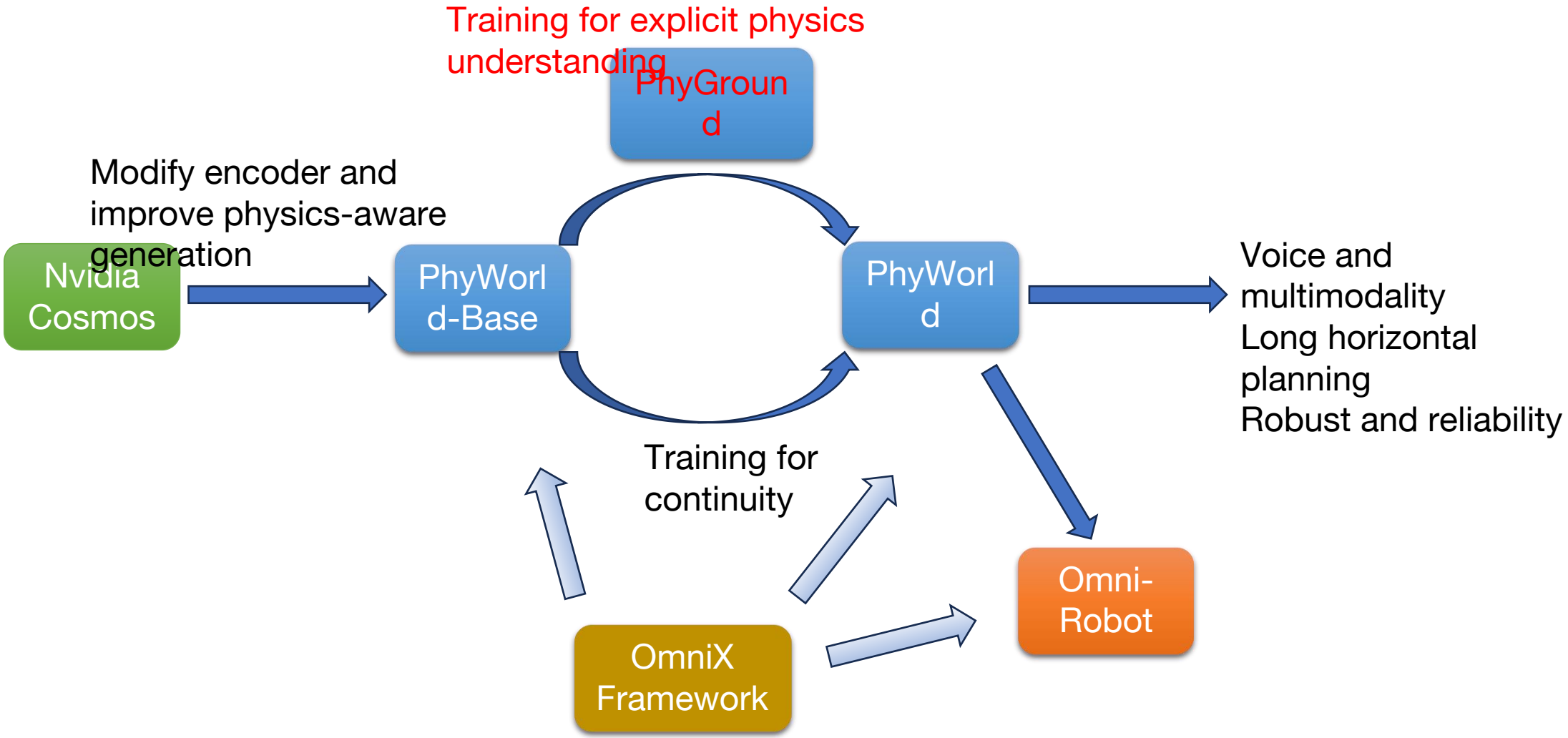
- Use Qwen3 VLM to filter whether video is sports/movement dense real world datasets.
- Use DeepSeek-VL2 to check again.

Train with Flow Matching

- Only continual training the DiT, freeze other parts



Our Innovations and Framework



PhyWorld Step II (Improve Physics-Understanding)

Training using DPO reinforcement learning

PhyGround: Physics Understanding Problem and Our Solution

- **Critical Evaluation Gaps:** Current benchmarks for evaluating physical understanding/reasoning are unreliable with three key limitations
  - Coarse Scores: 0/1 score
  - Human Bias: Small-scale annotation
  - Closed Judges: Non reproducible VLM judges
- Three Core Contributions of PhyGround:
  - **Fine-Grained Taxonomy**: 13 physical domains and associative laws
  - **Large-Scale Human Study**: Quality-controlled study with 352 annotators
  - **Open VLM Judge**: Reproducible PhyJudge-9B finetuned on human annotations

PhyGround: Physics Understanding Benchmarking Framework

C1: Criteria-Grounded Physics Taxonomy

3 Observable Domains

13 Per-Law Criteria

32 Sub-Questions

1–5 Likert Scale

Solid-Body | Fluid | Optical

Law 1 Score → Sub-Question 1, Sub-Question 2

Law 2 Score → Sub-Question 1, Sub-Question 2

⋮

Per-law + per-domain diagnostics

C2: Large-Scale, Quality-Controlled Human Evaluation

1. Large Annotator Pool

- 459 annotators
- 1,831 / 2,000 videos with ≥2 annotations
- 29.5K labels

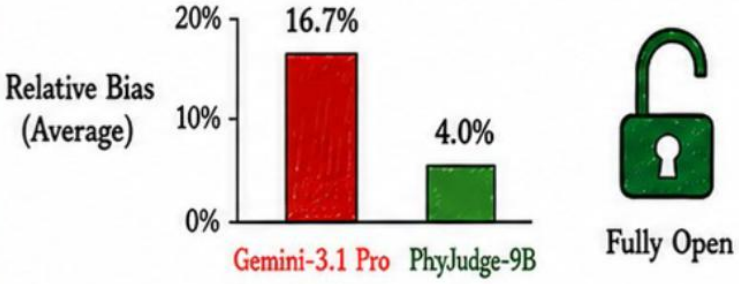
2. Social-Science Experiment Protocols

- Per-annotator session cap (reduces cognitive load)
- Annotator arrival and video order randomized
- Hypothesis-blind task framing

Split-half ranking $\rho > 0.90$

C3: PhyJudge-9B (Physics-Specialized VLM Judge)

Video Frames → Human Scores → PhyJudge-9B



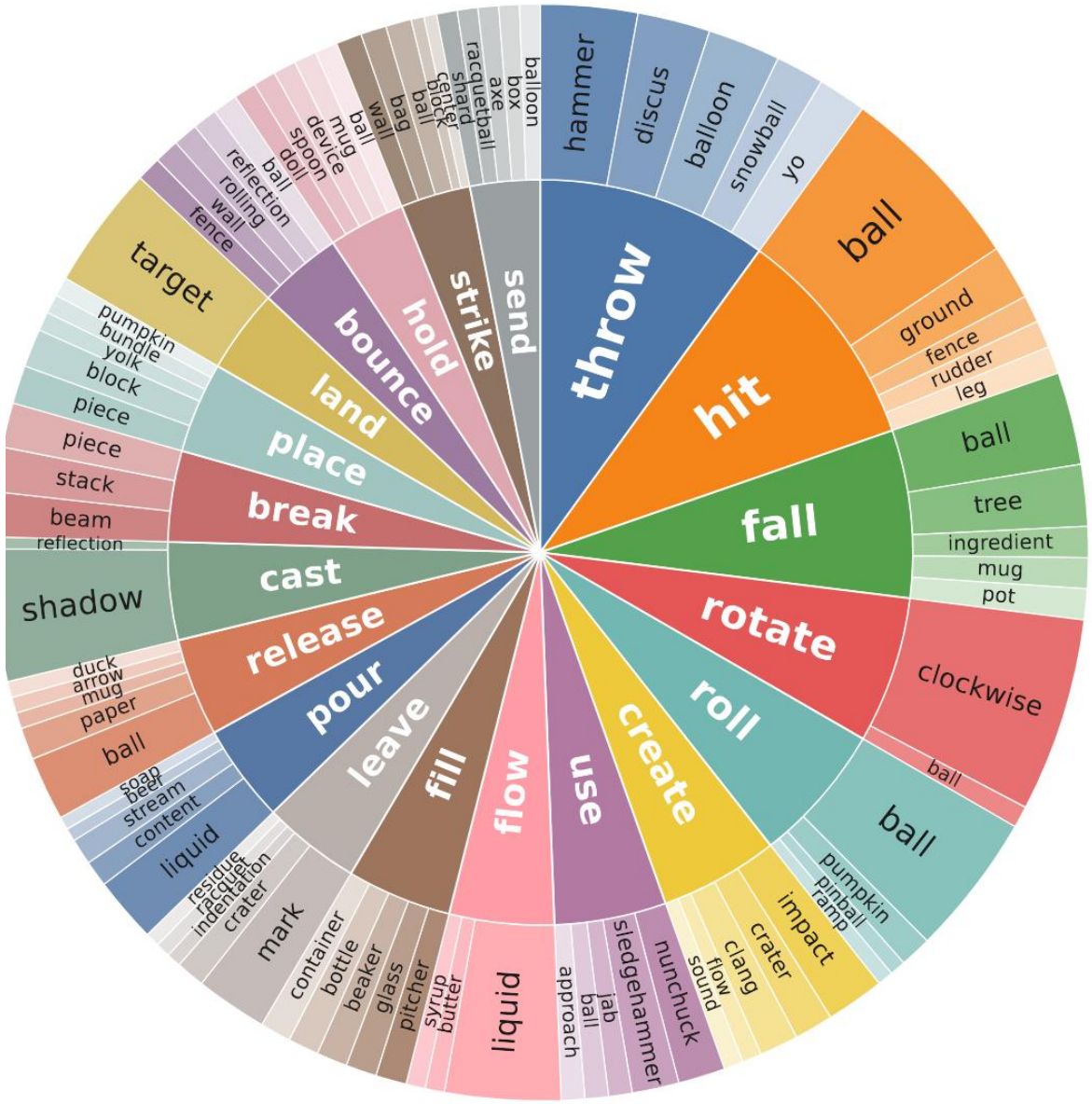
Open, low-bias VLM for reproducible physical evaluation

PhyGround: Physics Understanding Benchmarking Framework

Video related metrics

8 dimensions in total (7 VLM subjective + 1 objective computed):

Dimension	Score	Type
Semantic Alignment	1–5	VLM Subjective
Physical Plausibility (Overall)	1–5	VLM Subjective
Key Phenomenon Detection	0/1	VLM Subjective
Context/Background Naturalness	1–5	VLM Subjective
Physical Phase Temporal Ordering	1–5	VLM Subjective
Causal Order Accuracy	0/1	VLM Subjective
Frame-Wise Error (Motion Continuity)	0/1	Objective Computed
Object Persistence	1–5	VLM Subjective



PhyGround: Physics Understanding Benchmarking Framework

All 47 extreme disagreement cases (|ΔCDN| ≥ 3)

#1 cogvideox1.5-5b-i2v-wmb40



"Water flows over a small natural waterfall in a stream."

Gemini (no reasoning) 1 Gemini (forced reasoning) 1 QWEN 4

GEMINI'S REASONING:

The video exhibits a complete lack of dynamism. Despite the prompt describing flowing water over a waterfall, all frames are entirely static. There is no movement in the water, the ripples, or the surrounding vegetation. Because the expected physical motion is completely absent, the background physics and dynamism are severely broken.

QWEN'S RATIONALE:

CDN (Correct Dynamism / Background Physics):** The water flow looks realistic. Gravity pulls it down, turbulence forms at the bottom. The ripples propagate correctly downstream. There are no obvious glitches in the fluid dynamics visible at this resolution. Score: 4. 4. **PTV (Physical Temporal Verification):** The sequence of frames shows continuous flow. Time moves forward naturally. No weird backward motion or skipping. Score: 4. 5. **COA (Causal Order Accuracy):** Cause (water upstream) ->

#2 cosmos-predict2.5-14b-wmb40



"The stream gently flows over rocks and pebbles."

Gemini (no reasoning) 1 Gemini (forced reasoning) 1 QWEN 4

GEMINI'S REASONING:

The video is completely static across all frames. Despite the prompt describing a gently flowing stream, there is absolutely no movement in the water, rocks, or surrounding vegetation. Because the expected physical motion of flowing water is entirely absent, the dynamism is severely broken.

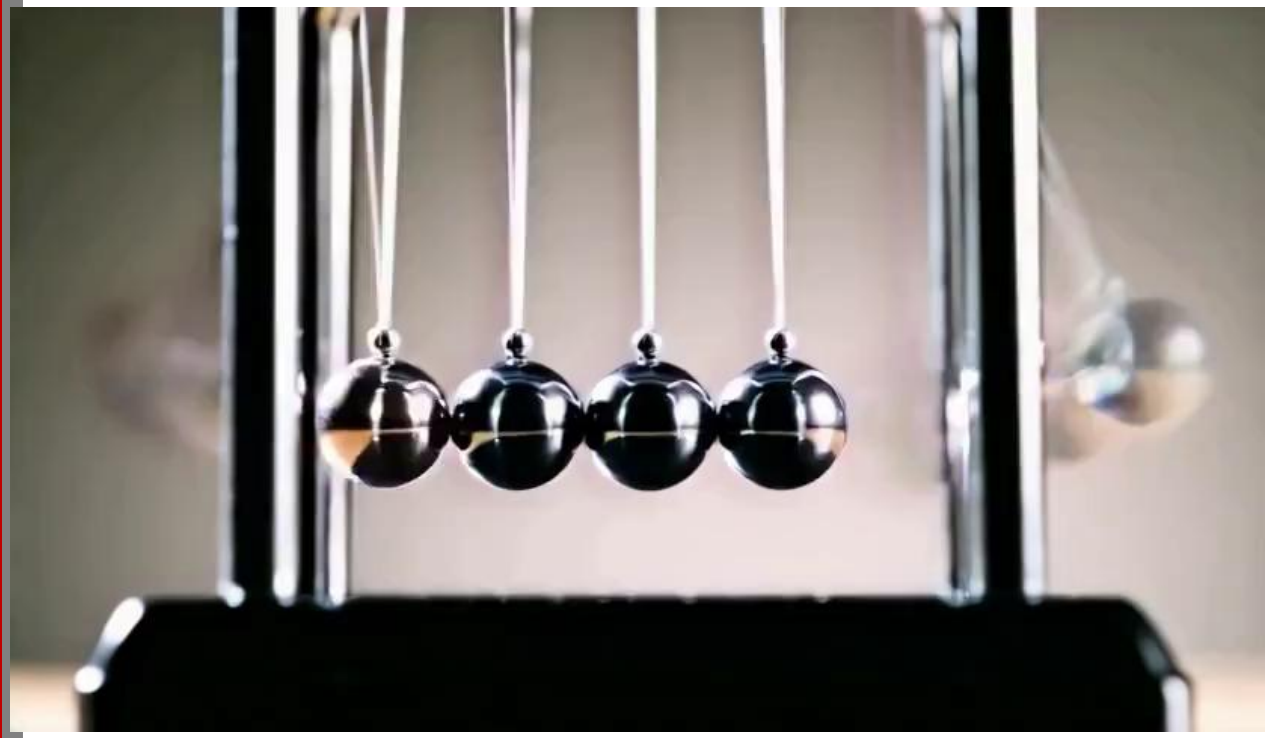
QWEN'S RATIONALE:

CDN (Correct Dynamism / Background Physics):** How realistic is the physics? - Water generally flows downhill. Here, the perspective suggests a downward slope. - The turbulence around the rocks looks plausible. - However, looking closely at the water movement, it seems a bit "boiling" or chaotic in a way that might be slightly exaggerated or

PhyGround: Human Evaluation Results

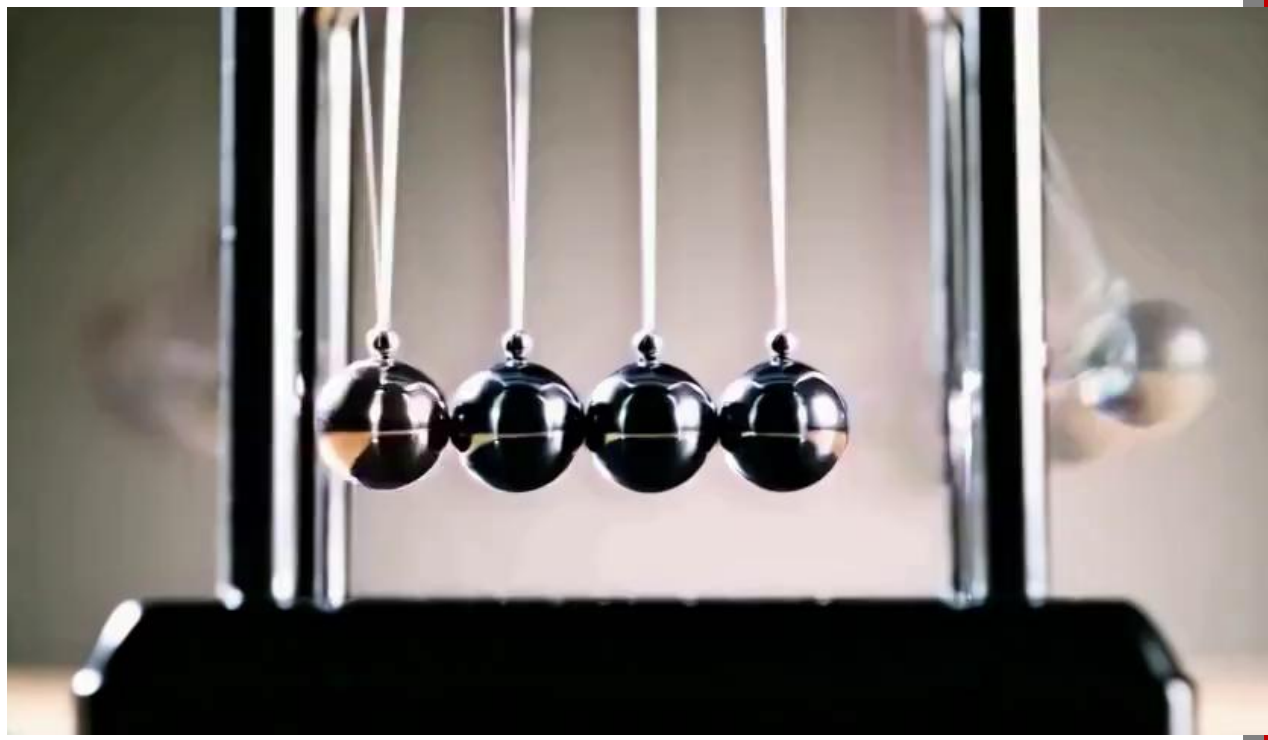
Model	General Physics ↑			Domain Physics ↑			Overall ↑
	SA	PTV	Persist.	Solid-Body	Fluid	Optical	
Wan2.2-27B-A14B	<u>3.10</u>	**3.37**	**3.50**	**3.23**	3.18	<u>3.55</u>	**3.28**
Veo-3.1†	**3.26**	<u>3.29</u>	<u>3.42</u>	<u>3.12</u>	**3.65**	**3.69**	**3.28**
OmniWeaving	2.97	3.17	3.34	2.98	<u>3.26</u>	3.22	3.10
Cosmos-14B	2.66	2.98	3.20	2.81	2.98	3.38	2.91
LTX-2.3-22B	2.58	2.74	2.72	2.62	3.07	2.98	2.69
Wan2.2-TI2V-5B	2.44	2.68	2.78	2.58	2.71	2.99	2.63
Cosmos-2B	2.33	2.56	2.77	2.53	2.77	3.35	2.58
LTX-2-19B	2.56	2.64	2.48	2.46	3.04	2.78	2.56

PhyWorld-Base vs. PhyWorld



PhyWorld-Base

- Color shift
- Speed change



PhyWorld

- Consistent color and speed

PhyWorld-Base vs. PhyWorld



PhyWorld-Base

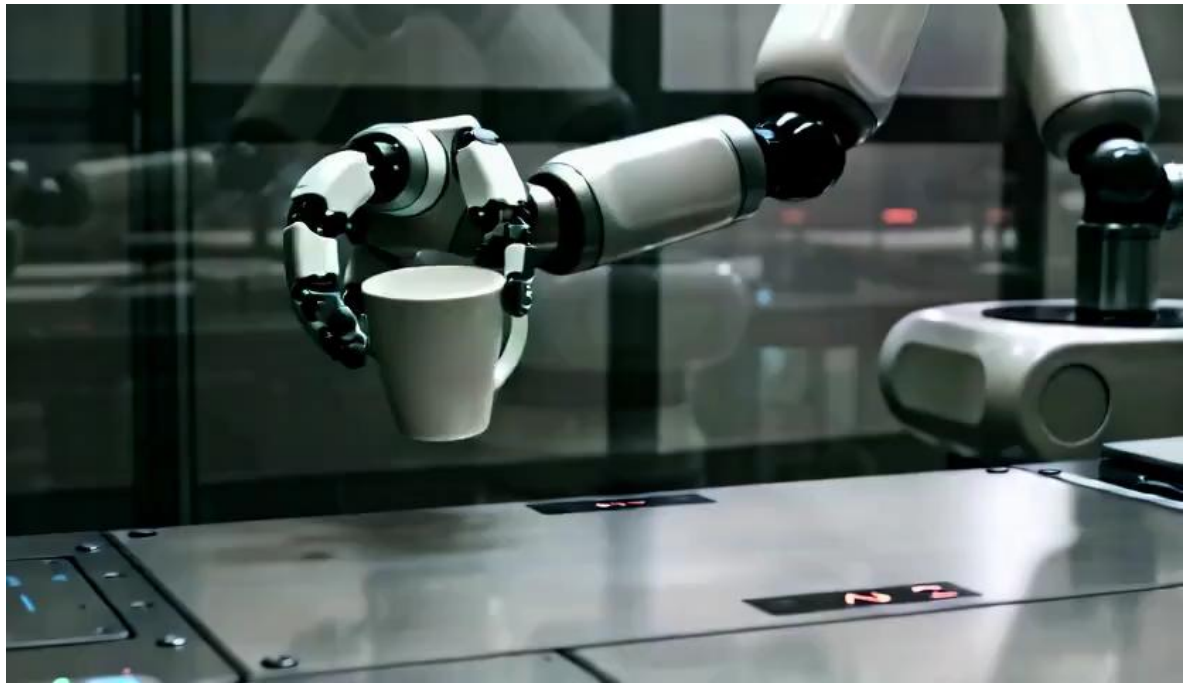
- Color shift (boxing gloves)
- Blur (fur)
- Exposure (light change)



PhyWorld

- Better consistency

PhyWorld-Base vs. PhyWorld



PhyWorld-Base

- Color shift
- Small violation of operation/physics



PhyWorld

- Consistent color and speed

Sample Evaluation Results

- Evaluation results on VBench
- Focusing on the video generation quality

Table 3: Evaluation results on VBench.

	subject consistency	background consistency	motion smoothness	dynamic degree	aesthetic quality	imaging quality	Avg.
Cosmos-2-2B	0.8874	0.9049	0.9638	0.5436	0.5239	0.6079	0.7386
OmniWeaving	0.9029	0.9068	0.9719	0.5562	0.5413	0.6208	0.75
LTX-2.3-22B	0.894	0.9183	0.9821	0.5491	0.5324	0.6259	0.7503
Cosmos-14B	0.8988	0.9228	0.9733	0.5586	0.5358	0.6285	0.7529
Wan2.2-I2V-A14B	0.9123	0.9275	0.9768	0.5538	0.5427	0.6224	0.7559
Ours	**0.9325**	**0.9437**	**0.9856**	**0.5641**	**0.5546**	**0.6322**	**0.7688**

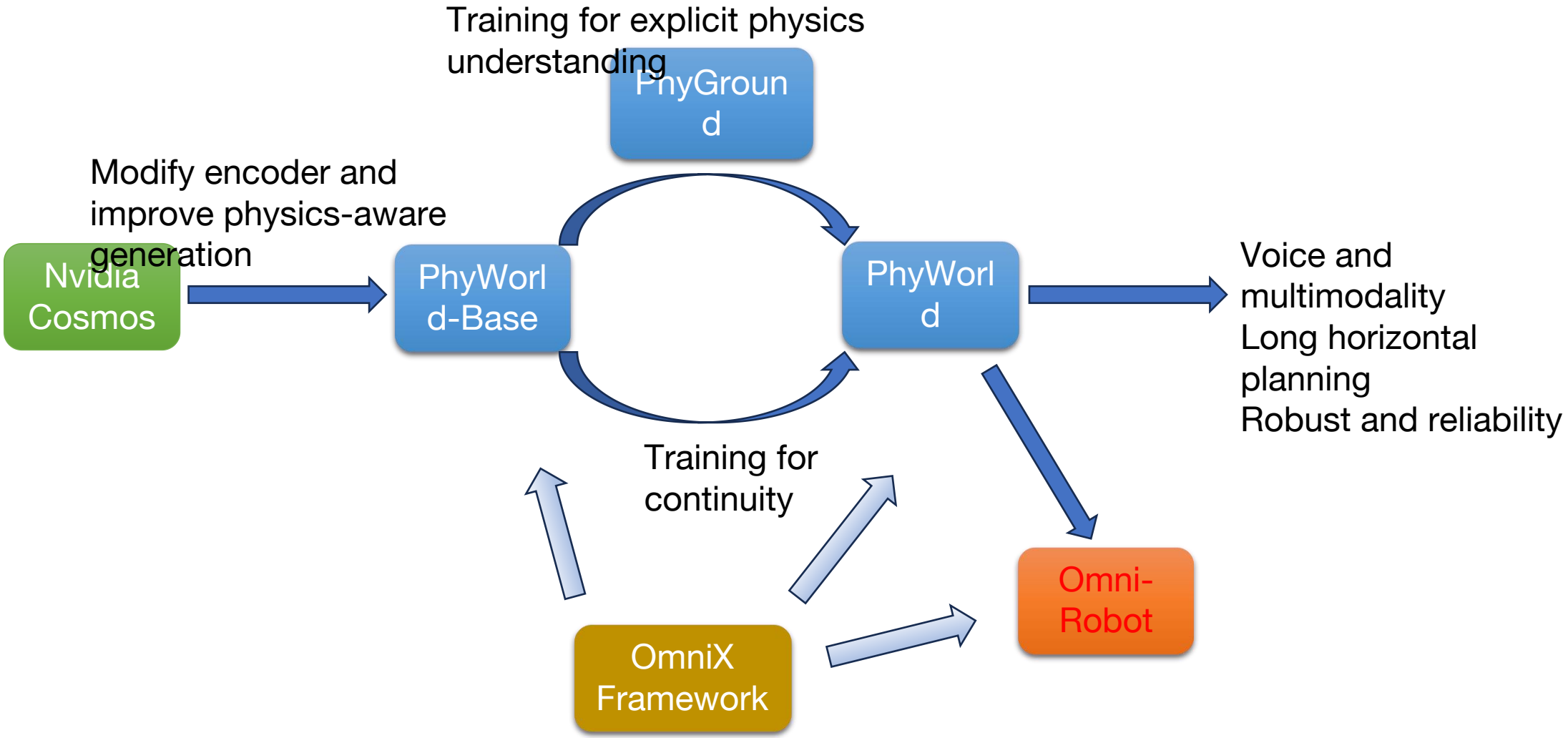
Sample Evaluation Results

- Evaluation results on our PhyGround benchmarking framework
- Focusing on the physical alignment

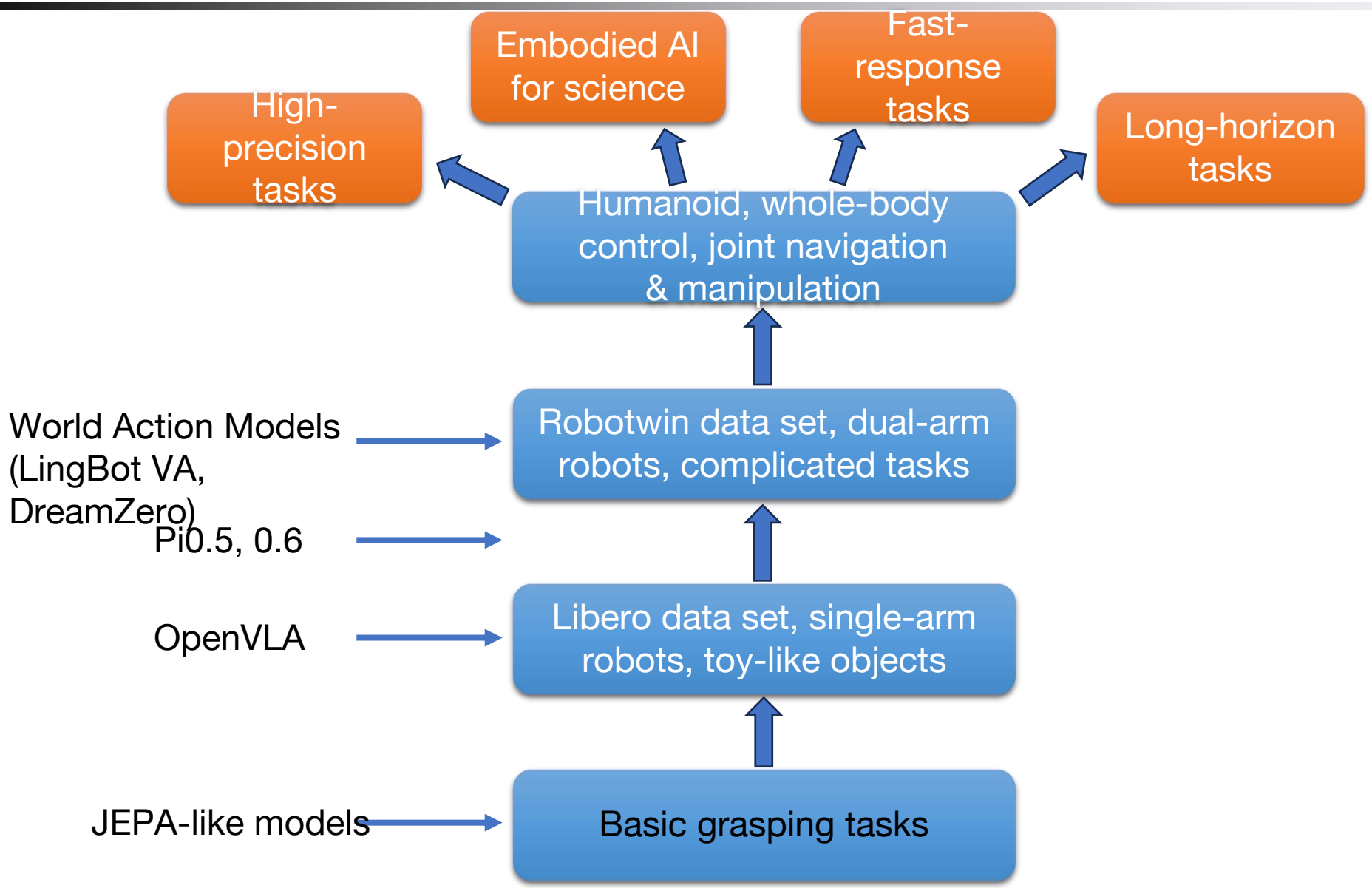
Table 4: Scores from our judge model on the 250-prompt humaneval set (1–5 scale, higher is better) for various models. SA, PTV, and Persist. are per-video general dimensions. Solid-Body, Fluid, and Optical are pooled over (video, law) units in each domain. Overall = 0.5 × mean(SA, PTV, Persist.) + 0.5 × pooled mean over (video, law) units across the three physics domains. **Bold** marks the best score per column.

	General Quality ↑			Physics Adherence ↑			
Model	SA	PTV	Persist.	Solid-Body	Fluid	Optical	Overall ↑
PhyWorldGen (ours)	**2.78**	**3.07**	**3.23**	**2.84**	**3.04**	**3.57**	**3.09**
Wan2.2-I2V-A14B	2.72	2.97	3.08	2.79	3.03	3.36	2.99
Cosmos-14B	2.60	2.73	3.07	2.72	2.92	3.53	2.80
OmniWeaving	2.68	2.73	2.92	2.71	2.99	3.13	2.78
LTX-2.3-22B	2.63	2.79	2.91	2.55	3.02	3.21	2.72
Wan2.2-TI2V-5B	2.48	2.70	2.76	2.61	3.01	3.45	2.68
LTX-2-19B	2.50	2.62	2.79	2.49	3.01	3.09	2.62

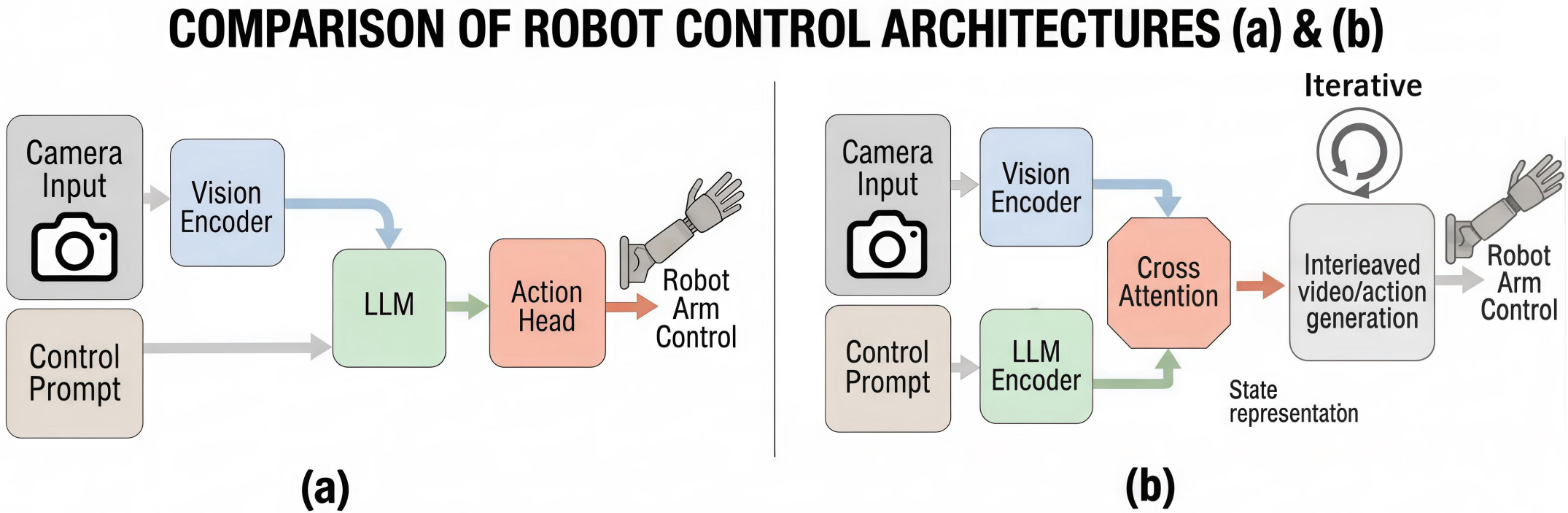
Our Innovations and Framework



Levels of Requirement in Embodied AI



What's Missing in the PI Models and VLA?



VLA: Robustness and generalization limitation

Lacks spatial information

WAM: Decouples representation and generation

DiT captures spatial information

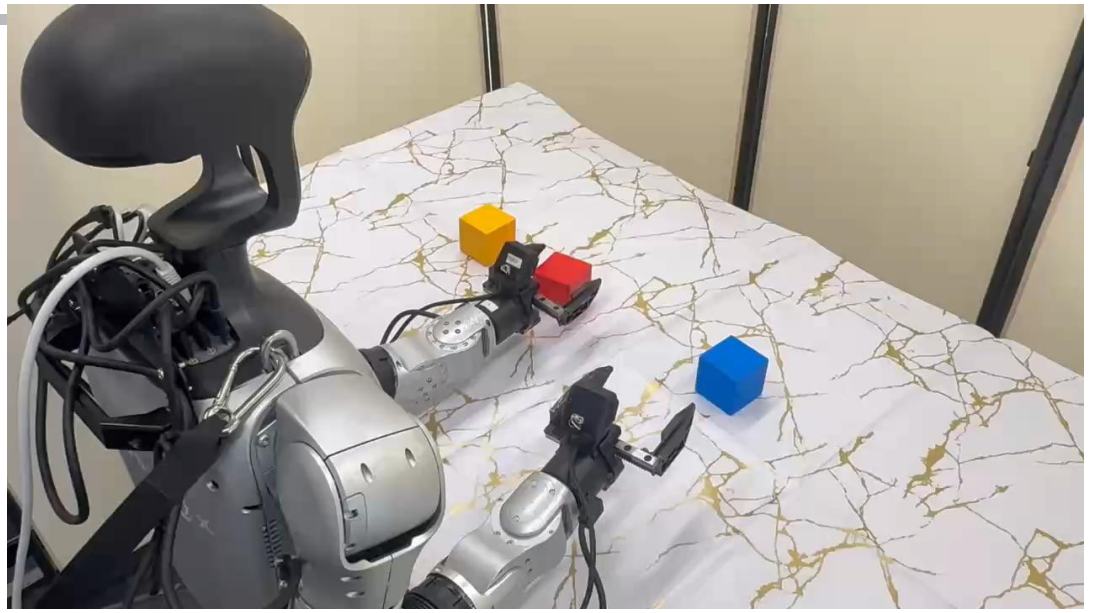
One Example: The Robustness Problem of VLA

Method: Pi 0.5

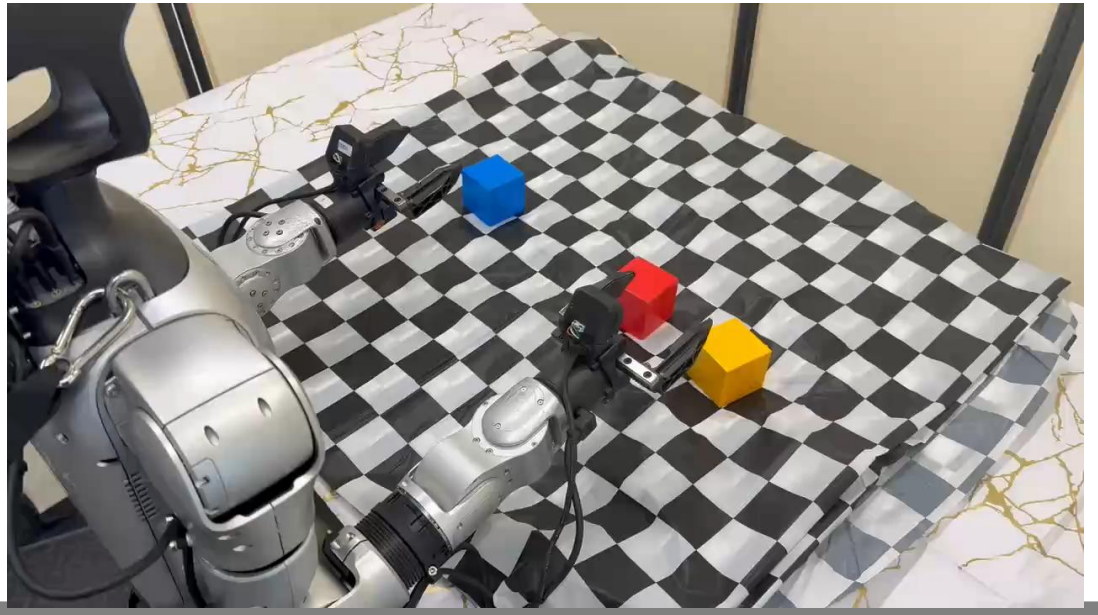
Key Finding: Background shift → **0% success**

- **White:** Stable
- **Marble / Checkered:** Less precise
- **Wood:** No targeting / grasping

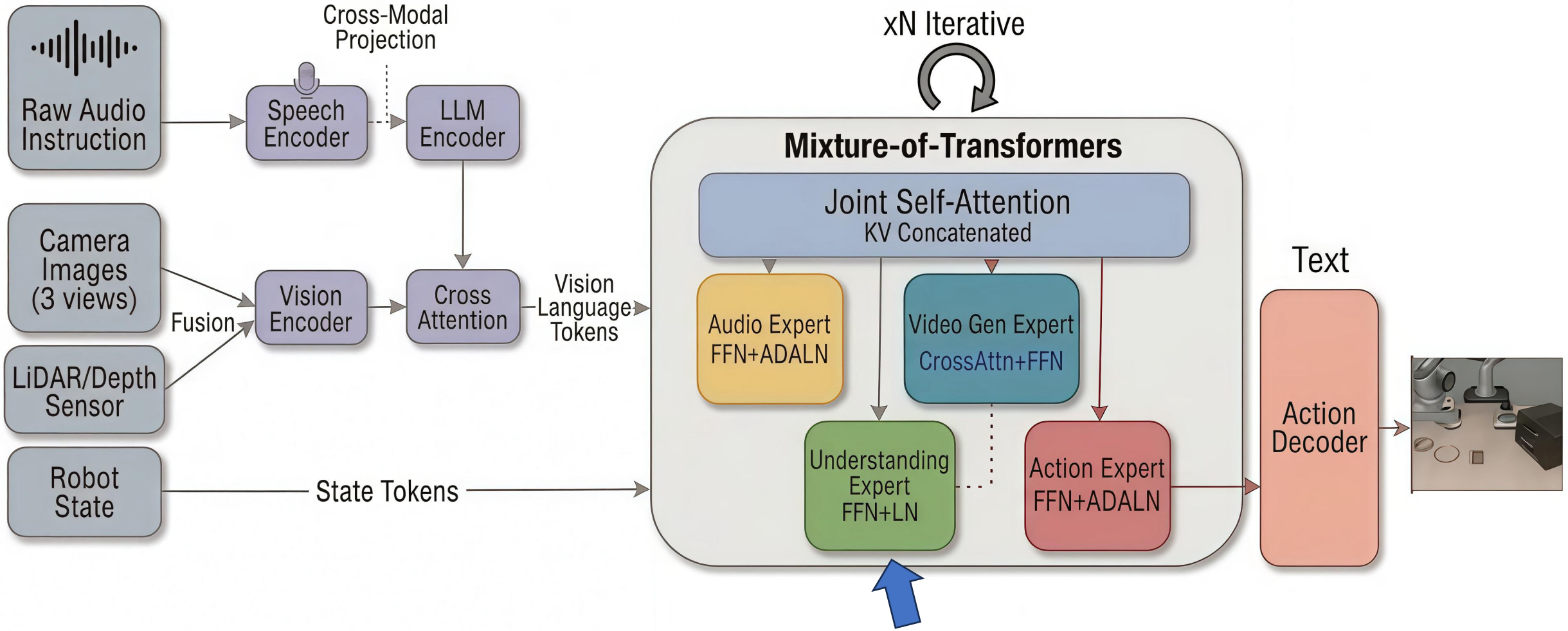
Takeaway: Wood texture likely breaks **object localization**.





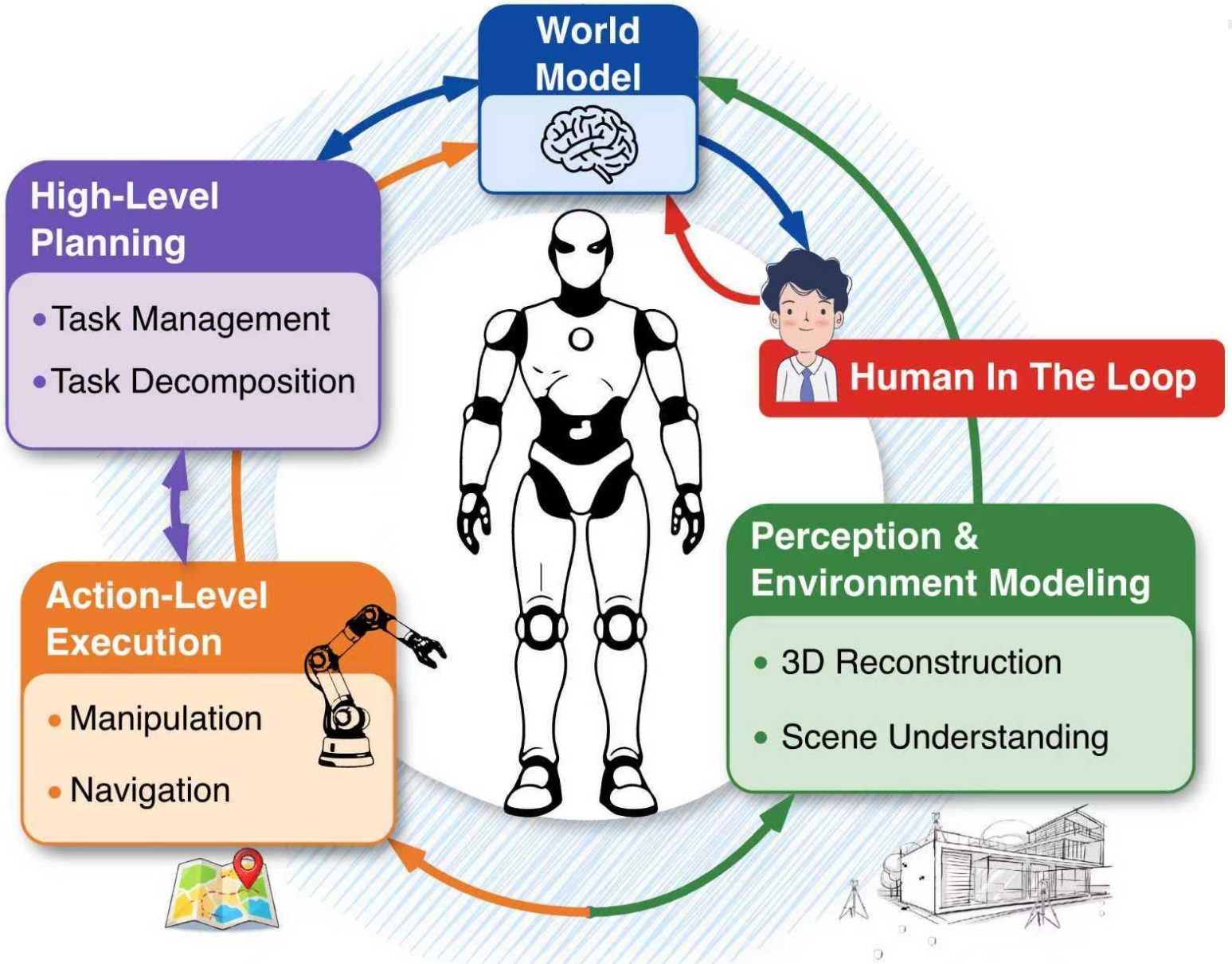


Enhanced World Model for Robot



The WAM should be able to understand the robot, object, and task

Omni-Robot: What can world model do in embodied AI

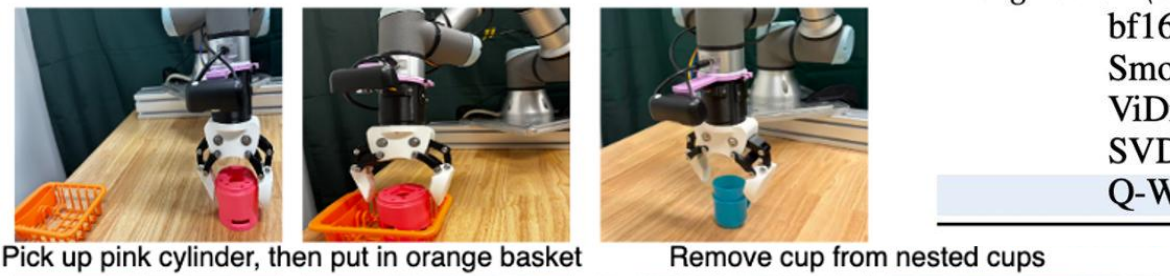


The "One World Model for All" Concept

Omni-Robot Acceleration: Quantization

Table 1: RoboTwin main results. BPW is the deployable weight cost; Mem is the memory of the targeted video and action blocks: their weights at the reported BPW plus the peak transient activation of one action chunk.

Model	Method	Precision	BPW	Clean	Randomized	Avg.	Mem (GB)
FastWAM (2-expert MoT, video generation backbone)							
	bf16 (upper bound)	W16A16	16.00	92.28	91.34	91.81	11.85
	SmoothQuant	W4A4	4.00	21.98	16.48	19.23	2.98
	ViDiT-Q	W4A4	4.41	82.00	80.48	81.24	3.61
	SVDQuant	W4A4	4.84	89.12	87.38	88.25	3.60
	Q-WAM (ours)	W4A4	**4.62**	**91.62**	**89.94**	**90.78**	3.44
ImageWAM (2-expert MoT, image editing backbone)							
	bf16 (upper bound)	W16A16	16.00	92.82	93.70	93.26	9.11
	SmoothQuant	W4A4	4.00	59.04	59.36	59.20	2.36
	ViDiT-Q	W4A4	4.76	78.76	82.08	80.42	3.05
	SVDQuant	W4A4	4.75	84.34	84.22	84.28	2.78
	Q-WAM (ours)	W4A4	**4.58**	**93.00**	**92.94**	**92.97**	2.69
LingBot-VA (shared backbone, not an MoT)							
	bf16 (upper bound)	W16A16	16.00	90.84	90.20	90.52	10.16
	SmoothQuant	W4A4	4.00	81.68	72.24	76.96	2.80
	ViDiT-Q	W4A4	4.41	80.82	— —	— —	3.18
	SVDQuant	W4A4	4.76	85.84	83.12	84.48	3.27
	Q-WAM (ours)	W4A4	**4.76**	**90.20**	**88.92**	**89.56**	3.27



(b) Real-world Task Examples

Nvidia Cosmos: Structure & Limitation

Inference latency, unit is ms, the lower the faster

Model	Hardware Platform	Official Speed	Our Optimized Speed (the same accuracy)
Pi-0.5	Nvidia Thor	268.0	31.7
Pi-0.5	RTX 5090	--	16.8
Omni-Robot	Nvidia Thor	456.7	80.2
Omni-Robot	RTX 5090	119.7	22.9

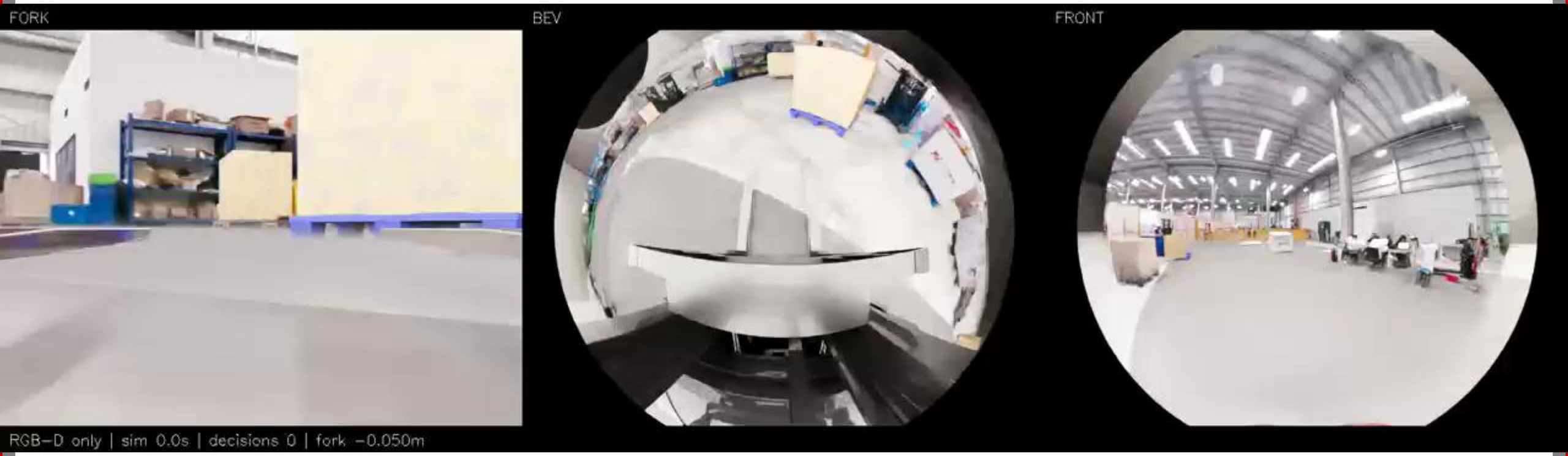
The first time to achieve real-time execution of WAM on the edge

Omni-Robot Demos (original speed)





Omni-Robot Demos (original speed)



Embodied AI for Science: Robotic Tasks for Optics Lab

A robot manipulation benchmark built on unmodified standard optomechanics and a real optical table

Objects **40 Thorlabs catalog parts in 8 categories**


Post


Post Holder


Base


Lens Tube


Mirror Mount


Lens Mount


Cage Plate


Stage

Task Levels **From single actions to routines to the full table**

L0 Perception

Optional

Part ID, hole coordinates

→

L1 Atomic

10 tasks

Pick, place, orient, insert

→

L2 Composite

6 tasks

Mount, relocate, swap

→

L3 Long-Horizon

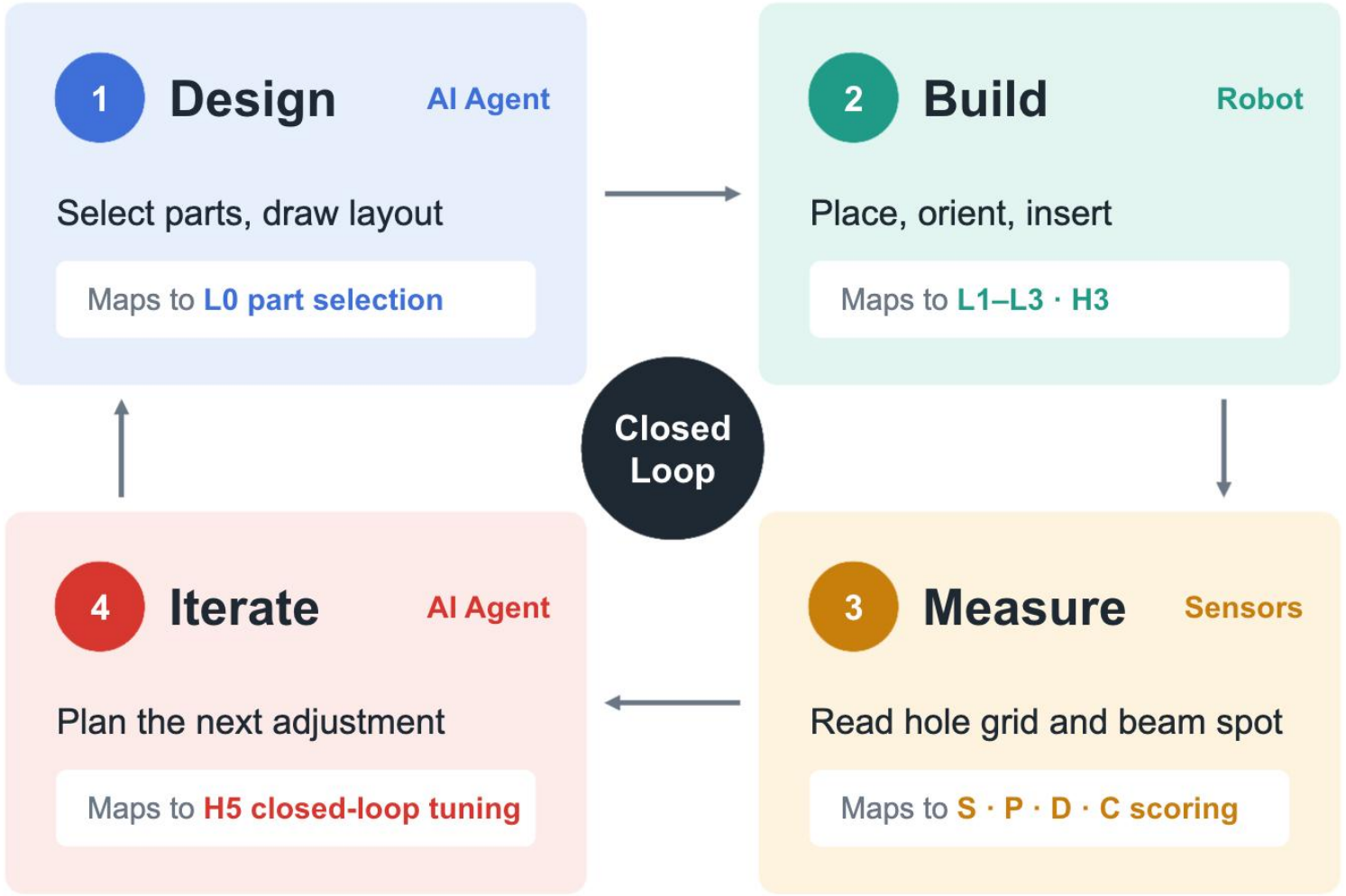
5 tasks

Tidy table, build from layout

L4 Generalization New parts, scenes, and instructions on L1–L3

Embodied AI for Science: From Benchmark to Auto Research

Optics experiments follow a design–build–measure–iterate loop; Optical Bench benchmarks every step



Why optics suits a closed loop

0.1° → 3.5 mm

Over 1 m, the beam itself is the ruler

Abbreviations

H3	Layout Build	3–6 elements from a top view
H5	Beam Check	Fold mirror puts beam on target
S	Success	Goal state held for 2 s
P	Precision	Hole, angle, height error
D	Disturbance	Aligned parts stay untouched
C	Compliance	No optic contact, no beam zone

Outreach Goal: Making GenAI Accessible to Everyone



Collaboration with industrial and academic partners for GenAI deployment community development

Contributing to open-source community of GenAI models & tools

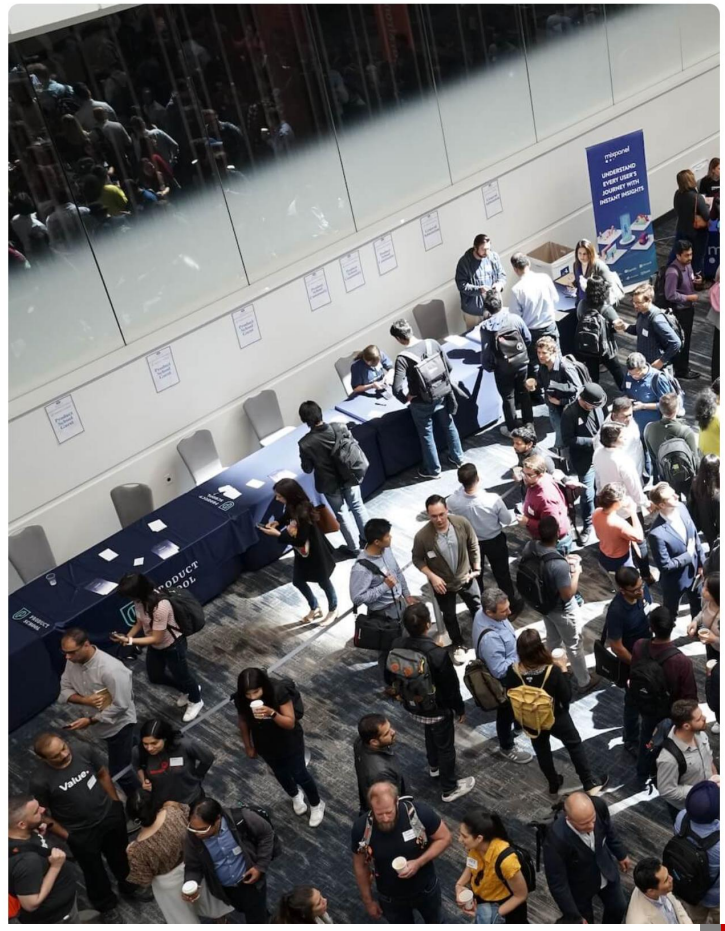
A Global Nexus for Open Source Innovation

The global tech landscape is replete with innovations, and much of its foundation is built upon the ethos of open source collaboration. At the forefront of championing this collaboration stands GOSIM, a platform dedicated to nurturing the brightest grassroots open-source projects from their inception to maturity.

GOSIM emerged from the shared vision of a global open-source community. Organized by and for the community, this entirely volunteer-based endeavor has one prime objective: to provide a stage where innovative open-source projects can shine, collaborate, and evolve. It's not just a platform; it's a movement. One that wholeheartedly embraces an open, diverse, and inclusive culture. If a project is global, innovative, grass roots ,and open-source, it has found its home at GOSIM.

- Organized by and created for the open-source community.
- Subscribes to an open, diverse, and inclusive culture.
- Volunteer based.

The Three Pillars of GOSIM



Thank You!

Watch our open-source projects

Northeastern University
College of Engineering